Chapter 5
Mathematical Interlude



No truth seems true.

Maya Agelou, Poet, 1976

We are now in the second part of the book, covering the material of *Quantum Operator Mechanics*, which spans Chaps. 5–12. We have a video that provides an overview of this second part https://tinyurl.com/bdeu6de9 and of this chapter https://tinyurl.com/mv4za5by.

5.1 Devious Math Tricks

Remember that kid from algebra class who sat across from you and seemed to know the answer to every question faster and more accurately than you? I am going to let you in on a secret—most likely that curve buster was not a genius, nor born to do math—instead, they simply learned a number of devious tricks to get through the material faster and with fewer errors. After finishing this section, you will be able to do so too!

The biggest myth to dispel is that geniuses in algebra have photographic memories and can recall the relevant formulas and apply them at will. Such people may exist, but the braniac across the row from you was not one of these. The reality is you only need to know about ten important equations, and by using them as I am about to describe to you, you will find that you will just know them when you need them. You too can become an algebra master!

Devious trick number 1: $x + 0 = x$.

This is called the "add-zero" trick. It comes up so often that you cannot get through most problems without encountering it. The problem is no one has told you about it, so you often stumble onto it and fail to see its importance. We will see and use the add-zero trick again and again. Right now, let me illustrate it to you. Consider the following problem: Factor $x^2 - 25$.

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J. K. Freericks, *Quantum Mechanics Done Right*,
https://doi.org/10.1007/978-3-031-87845-9_5

Go ahead and give it a try, but do not just write down the answer if you have memorized it, and see if you can derive it with the add-zero trick.

Here is how it works. An educated guess suggests we should add a number times x as one of the terms, and the 25 suggests that we add $5x - 5x$ to obtain

$$\begin{aligned} x^2 + \underbrace{5x - 5x}_{=0} - 25 &= (x^2 + 5x) - (5x + 25) \\ &= x(x+5) - 5(x+5) \\ &= (x-5)(x+5). \end{aligned} \tag{5.1}$$

All we did was add zero! The creativity was to recognize what zero to add. Indeed, this is how you apply the technique. You will quickly learn what ideas might work for what to add by using the method multiple times. One strategy to follow is to look for a term that is missing in a familiar identity. When this occurs, just add and subtract the missing term and see if using the add-zero identity helps!

Devious trick number 2: $x \cdot 1 = x$.

Just like the "add-zero" trick, the "multiply-by-one" trick can be used to simplify many different expressions. It is often seen in cases where we want to rationalize a denominator, such as the following. Rationalize the denominator of $(1 - \sqrt{3})/(1 + 2\sqrt{3})$.

The strategy is to multiply the numerator and the denominator by the same factor such that the denominator no longer has a square root. Recalling our work above can be written more generally as $a^2 - b^2 = (a+b)(a-b)$ yields the strategy to use $1 = (1 - 2\sqrt{3})/(1 - 2\sqrt{3})$ as the term we multiply by. Now, compute

$$\frac{1-\sqrt{3}}{1+2\sqrt{3}} \times \frac{1-2\sqrt{3}}{1-2\sqrt{3}} = \frac{1-\sqrt{3}-2\sqrt{3}+6}{1-12} = -\frac{7-3\sqrt{3}}{11}. \tag{5.2}$$

This multiply-by-one trick can also be used to simplify products of integers via factorials. Recall the factorial symbol $n! = 1 \times 2 \times 3 \times 4 \cdots \times (n-1) \times n$ for the product of the first n nonzero integers. Then, the following product can be rearranged in terms of factorials as follows:

$$\begin{aligned} (n+1)(n+2)\cdots(2n-1)(2n) &= \frac{1 \cdot 2 \cdot 3 \cdots n}{1 \cdot 2 \cdot 3 \cdots n} \times (n+1)(n+2)\cdots(2n-1)(2n) \\ &= \frac{(2n)!}{n!}. \end{aligned} \tag{5.3}$$

Devious trick number 3: Remember simple identities and use them to construct more complicated identities.

As I mentioned before, there are only a handful of identities worth remembering. Nearly everything else can be derived from these few. As an example, think about

trigonometric identities. There are loads and loads of them. But they all follow from two main results—remembering the definitions of trigonometric functions in terms of sine and cosine and using the fact that $\cos^2\theta + \sin^2\theta = 1$. Let us illustrate how.

Suppose I want to express $\sec^2\theta$ in terms of $\tan\theta$. I simply recall that $\sec\theta = 1/\cos\theta$. Then, I use the multiply-by-one trick with $1 = \cos^2\theta + \sin^2\theta$ to find that

$$\begin{aligned}\sec^2\theta &= \frac{1}{\cos^2\theta} = \frac{1}{\cos^2\theta} \times (\cos^2\theta + \sin^2\theta) \\ &= \frac{\cos^2\theta}{\cos^2\theta} + \frac{\sin^2\theta}{\cos^2\theta} = 1 + \tan^2\theta. \end{aligned} \tag{5.4}$$

Devious trick number 4: Recognize simple identities, when expressed in a complex form.

This devious trick takes a bit more practice to master, as it requires you to use abstraction to recognize the underlying simple identity. Here is an example. We already saw that $x^2 - 25 = (x+5)(x-5)$. We should be able to recognize this in different contexts, where x is more complicated. For example, consider

$$\frac{\sin^2\theta - 25}{\cos^2\theta} = \tan^2\theta - 25\sec^2\theta = (\tan\theta + 5\sec\theta)(\tan\theta - 5\sec\theta). \tag{5.5}$$

You can see the relationship between these two identities rather clearly.

A more complicated example is to factor $x^4 + 2x^2 - 24$. To do this, we first use the add-zero trick and then recognize our previous identity. This is done as follows:

$$\begin{aligned} x^4 + 2x^2 + \underbrace{1-1}_{=0} - 24 &= (x^2+1)^2 - 25 \\ &= (x^2 + 1 + 5)(x^2 + 1 - 5) = (x^2+6)(x^2-4). \end{aligned} \tag{5.6}$$

Note how we had to recognize that $x^4 + 2x^2 + 1 = (x^2+1)^2$ was the square in the expression. To build up practice in doing this, you might find it easier to first let $x^2 = y$ and then factorize $y^2 + 2y - 24$, and then, let $y \to x^2$ at the end.

Devious trick number 5: Use only the information that you know.

This trick also has an added point—learn how to make small changes to what you know to lead to what you want to derive. This trick also takes time and practice to recognize it, but it is worthwhile doing so, as it is a very important trick. We will illustrate it by example. Consider evaluating the following sum:

$$x^2 + x^4 + x^6 + \cdots = \sum_{n=1}^{\infty} x^{2n}. \tag{5.7}$$

Note that the right-hand side is our summation notation. You think of it as a rule for adding numbers. Start with $n = 1$, which gives x^2, and then, add to that the

result with $n = 2$, which gives x^4, and then, add $n = 3$, which gives x^6, and so on, without ever stopping. This sum looks like our friend the geometric series, but it is with respect to x^2 and it is missing the first term. The idea here is to use the geometric series that you know to solve this. One way is by using add zero:

$$\sum_{n=1}^{\infty} x^{2n} = \sum_{n=0}^{\infty} x^{2n} - 1 = \frac{1}{1-x^2} - 1 = \frac{x^2}{1-x^2}. \tag{5.8}$$

We could also proceed another way, by recognizing that

$$\begin{aligned} x^2 + x^4 + x^6 + \cdots &= x^2 \underbrace{(1 + x^2 + x^4 + \cdots)}_{\text{geometric series}} \\ &= x^2 \sum_{n=0}^{\infty} x^{2n} = x^2 \frac{1}{1-x^2} = \frac{x^2}{1-x^2}, \end{aligned} \tag{5.9}$$

as before. Note how we took what we knew, modified it slightly, and discovered something new. I always think about this trick when I think about a scene from Douglas Adams, *The long dark tea-time of the soul*, where the holistic detective Dirk Gently has a curious way of investigating. An example is when he is looking for a woman who lives at a particular address in a town he does not know. Rather than looking up the address, he just gets in the car and starts driving. The car gets into an accident, and lo and behold, the woman he was trying to find was the driver of the other car. He says, "I may not have gone where I intended to go, but I think I have ended up where I needed to be" Adams (1989). Using what you know is similar. Just start from what you know and try to get to where you want to be. *Learning to use what you know, and only what you know will help you a lot in learning quantum mechanics.*

We end by discussing one other technical skill, which involves shifting indices in summations. Students often struggle with this skill. Yet it is straightforward to master it. Doing so takes you to the next level of algebraic skill. Let us look at the last example in Eq. (5.7). We know the geometric series starts with $n = 0$ and not $n = 1$. So, let us shift $n \to \bar{n} + 1$. Then, $\bar{n}$ runs from 0 to infinity (because n ran from 1 to infinity). We obtain

$$\sum_{n=1}^{\infty} x^{2n} = \sum_{\bar{n}=0}^{\infty} x^{2\bar{n}+2} = \sum_{\bar{n}=0}^{\infty} x^{2\bar{n}} \cdot x^2 = \frac{1}{1-x^2} x^2 = \frac{x^2}{1-x^2}, \tag{5.10}$$

just like before. We will see this index-shifting skill becomes critically important in mastering many identities we develop later. I will help you recognize and develop this skill as we move forward.

Table 5.1 Ten mathematical identities you need to know

Identity	Formula
The quadratic equation	$ax^2 + bx + c = 0$ has two roots $x_\pm = \frac{-b \pm \sqrt{b^2-4ac}}{2a}$
Fundamental trigonometric identity	$\cos^2\theta + \sin^2\theta = 1$
Trigonometric definitions	$\sec\theta = \frac{1}{\cos\theta}$, $\tan\theta = \frac{\sin\theta}{\cos\theta}$, $\text{cosec}\theta = \frac{1}{\sin\theta}$, $\text{cotan}\theta = \frac{\cos\theta}{\sin\theta}$
Double-angle formulas	$\cos 2\theta = \cos^2\theta - \sin^2\theta$, $\sin 2\theta = 2\sin\theta\cos\theta$
Factorization of the difference of squares	$a^2 - b^2 = (a+b)(a-b)$
Geometric series	$\sum_{n=0}^{\infty} x^n = \frac{1}{1-x}$ for $\lvert x\rvert < 1$, $\sum_{n=0}^{N} x^n = \frac{1-x^{N+1}}{1-x}$
Binomial theorem	$(x+y)^N = \sum_{n=0}^{N} \binom{N}{n} x^{N-n} y^n = \sum_{n=0}^{N} \frac{N!}{(N-n)!n!} x^{N-n} y^n$
Power series for the exponential	$e^x = \sum_{n=0}^{\infty} \frac{1}{n!} x^n$
Relation of trig functions to exponentials	$\cos\theta = \frac{1}{2}(e^{ix} + e^{-ix})$, $\sin\theta = \frac{1}{2i}(e^{ix} - e^{-ix})$
Relation of hyp functions to exponentials	$\cosh x = \frac{1}{2}(e^x + e^{-x})$, $\sinh x = \frac{1}{2}(e^x - e^{-x})$

We end this section with ten important identities that you need to know to work through this book. We will use them freely. We will develop some of these in this chapter if you have never seen them before. They are summarized in Table 5.1. The video Devious Math Tricks https://tinyurl.com/4k4wjxv7 summarizes this discussion.

Section 5.1 Problems:

There are nine problems to practice your devious tricks. Beware, some of them are challenging.

5.1.1 Factoring

Factor $x^2 + 2xy + y^2 - 9$.

5.1.2 Double-Angle Formula

Use the double-angle formula in the form $\cos\theta = \cos^2\frac{\theta}{2} - \sin^2\frac{\theta}{2}$ to show that $\cos\frac{\theta}{2} = \pm\sqrt{\frac{1+\cos\theta}{2}}$. How do we choose which sign is correct?

5.1.3 Half-Angle Formula

Use the same double-angle formula to find $\sin\frac{\theta}{2}$.

5.1.4 Power Series for Exponentials

Show that $\sum_{n=0}^{\infty}\frac{x^n}{(n+1)!} = \frac{e^x-1}{x}$.

5.1.5 Add Zero Instead of Factorize

Show that $x^2 = 3+2\sqrt{2}$ implies $x = \pm\sqrt{1+\sqrt{2}}$ by using the "add-zero" trick (do not use the quadratic formula).

5.1.6 Double Trouble

Use the double-angle formulas for sine and cosine to show that $\tan 2\theta = \frac{2\tan\theta}{1-\tan^2\theta}$.

5.1.7 Geometric Series

Use the finite geometric series formula (in our list of ten formulas) to show that $1+\sqrt[4]{2}+\sqrt{2}+\sqrt[4]{8} = \frac{1}{\sqrt{2}-1}$.

5.1.8 Fun with Factorials

Show that $\frac{(2n)!}{2^n n!} = 1\cdot 3\cdot 5\cdots(2n-3)\cdot(2n-1) = (2n-1)!!$, where we define the double factorial symbol to mean skip every other term in the product of integers. An

even double factorial is a product of all even integers, while an odd double factorial is a product of all odd integers.
Hint: think of how many even integers are in $(2n)!$. How can they be removed?

5.1.9 Trigonometric Summation

Show $\sum_{n=1}^{\infty}(\cos\theta)^{2n} = \frac{1}{\tan^2\theta}$, for $|\theta| < \frac{\pi}{2}$.
Hint: use the geometric series.

5.2 Complex Numbers

Up until now, we have used arrows to represent quantum probability amplitudes. Turns out, those arrows are closely related to complex numbers, and we need to understand complex numbers well to make progress in more sophisticated quantum mechanics calculations. So, we now delve into the world of complex numbers.

You probably already learned that $\sqrt{-1}$ is not a real number but is an imaginary number called i, and we can form complex numbers $z = x + iy$ from these real and imaginary numbers. Complex numbers often appear imposing and foreign, but as you work with them more and more, you will find they become familiar and comfortable. We call x the real part of z ($x = \mathrm{Re}\, z$) and y the imaginary part of z ($y = \mathrm{Im}\, z$); note that *both* x and y are real numbers.

You should already be familiar with working with real numbers in a plane, denoted by an ordered list of two real numbers (x, y). The real plane has both magnitudes and directions to it. We often use polar coordinates of r and θ to denote the radial distance from the origin (at (0,0)) and the angle measured counterclockwise from the horizontal (x) axis. In this context, points on the plane can be thought of as vectors. Complex numbers are also two-dimensional objects that can be represented with an ordered list $x + iy \rightarrow (x, y)$, and we refer to the points as lying in what we now call *the complex plane*. Here, we can still think of representing the points on the complex plane in terms of vectors, but the big difference from the two-dimensional real plane, and the complex plane, is for complex numbers, we have a rule that allows us to multiply two complex numbers to produce a third complex number, the product of the two. We just use the fact that $i^2 = -1$, so that $(x+iy)(x'+iy') = xx'-yy'+i(xy'+yx')$. This difference means we can do much more with complex numbers than we can with a two-dimensional real plane, as you will see in the discussion below.

As an example, we plot the complex number $z = 2 + 3i$ in Fig. 5.1, to illustrate both the Cartesian representation and the polar representation. The real part $x = 2$ and the imaginary part $y = 3$ are the Cartesian representation of the complex number. The polar representation used the radius $r = \sqrt{x^2 + y^2}$ (here given by $r = \sqrt{2^2 + 3^2} = \sqrt{13}$) and the polar angle (measured from the horizontal axis) $\theta = \tan^{-1}(y/x)$ (here given by $\theta = \tan^{-1}(3/2) \approx 56.3^{\circ}$); one has to shift the

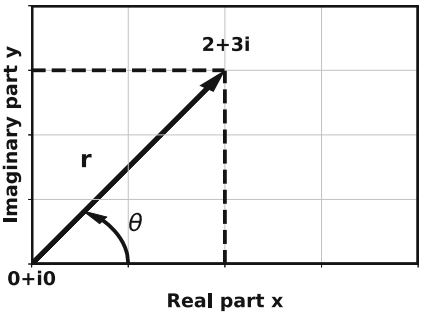


Fig. 5.1 Comparison between Cartesian and polar representations of complex numbers. The complex number $z = 2 + 3i$ can be represented as a Cartesian element (2, 3) or as a polar element $r = \sqrt{2^2 + 3^2}$ and $\theta = \tan^{-1}\left(\frac{3}{2}\right)$. The origin $0 + i0$ is denoted by (0, 0) or $r = 0$

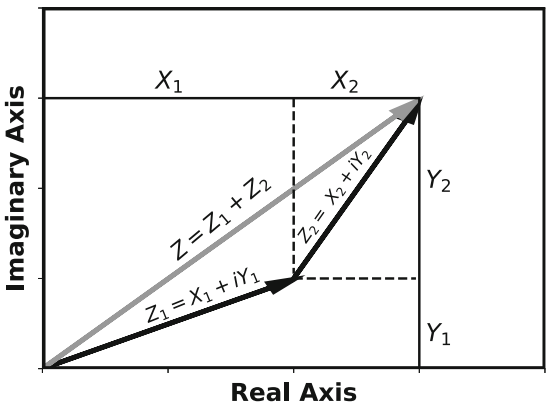


Fig. 5.2 Schematic of how we add two complex numbers by hooking head to tail and drawing a final arrow. Note how this is the same as adding the real parts for the real part of the sum and adding imaginary parts for the imaginary part of the sum

arctangent by $\pm\pi/2$ for complex numbers with $y < 0$. These coordinates represent the distance and angle that a given complex number has relative to the origin (and the x-axis). Of course, which coordinates you choose to use, Cartesian or polar is a matter of taste. Some calculations are easier in one form (addition in the Cartesian form), and others are easier in the other form (multiplication in the polar form). So, we need to gain practice in using *both* forms and be able to convert from one form to another.

If we want to add two complex numbers, the operation is identical to how we add vectors in a two-dimensional plane—we just add the horizontal components and the vertical components. Multiplying complex numbers is something new. The only multiplication with vectors was the dot product or the cross product, neither of which is what we want for multiplication of complex numbers, because we want the product to be a complex number as well. The rule for multiplying complex numbers is most easily stated in polar form—we multiply the radii together and we add the polar angles. This is one of the major differences between complex numbers and the real two-dimensional plane.

Now, let us think about the arrows we have been using. For the arrows, we added them by hooking the head to tail and then finding the final arrow. As Fig. 5.2 shows, this is identical to adding numbers in the plane. It turns out we do exactly the same

thing to add complex numbers. If $z_1 = x_1 + iy_1$ and $z_2 = x_2 + iy_2$, then the sum is $z_1 + z_2 = (x_1 + x_2) + i(y_1 + y_2)$. For example, add $z_1 = 4 + 7i$ to $z_2 = 3 - 2i$ and see that you obtain $7 + 5i$ for the sum. Recall as well that we also had a procedure for multiplying arrows. We multiplied their lengths and added the hours on the clocks. This is the same as saying the radius of $z_1 \times z_2$ is $r_1 \times r_2$ and the polar angle for $z_1 \times z_2$ is $\theta_1 + \theta_2$. *Hence, the arrows we have been using all along are just complex numbers!* We find the Cartesian form is the best to use when adding them together and the polar form is the best to use when multiplying them. It turns out that the rules we need to follow for working with complex numbers are simple and natural—in particular, we now can add two complex numbers and obtain a complex number as the sum, and we can multiply two complex numbers and obtain a complex number as the product. There is one issue in relating arrows to complex numbers that we do need to address. In polar form, the angle is measured in a counterclockwise direction and from the horizontal axis, while for the arrows, we measured time in a clockwise direction with respect to the vertical axis. The issue of clockwise versus counterclockwise will become clear when we discuss time evolution (they are indeed opposite for good reason). The measurement with respect to the horizontal or vertical axes is a convention and they are just different.

Complex numbers have additional important operations on them, which we will now discuss. We start from the notion of a complex conjugate, denoted with an asterisk superscript: $z^* = (x + iy)^* = x - iy$. It is defined as a reflection of the complex number about the horizontal axis, which changes the sign of the imaginary part. This is illustrated in Fig. 5.3. This rule is simple to implement. Anytime you have an i in a complex number, you change $i \to -i$ for the complex conjugate—said differently, you change the sign of the coefficient of i. Hence, the complex conjugate of a complex number is the original complex number or $(z^*)^* = z$. The proof is direct—$z = x + iy$ implies that $z^* = x - iy$, which implies $(z^*)^* = x + iy$. You can also show this by noting that two reflections in the same mirror returns the original complex number.

The conjugate comes in handy in computing the radius in polar form. We denote the radius by the norm (or absolute value) of z, $|z| = r$. Since $r = \sqrt{x^2 + y^2}$, we have $|z|^2 = r^2 = x^2 + y^2 = zz^*$. Proof: $zz^* = (x + iy)(x - iy) = x^2 + ixy - ixy + i(-i)y^2 = x^2 + y^2$, since $i(-i) = -i^2 = -(-1) = 1$. Note that this

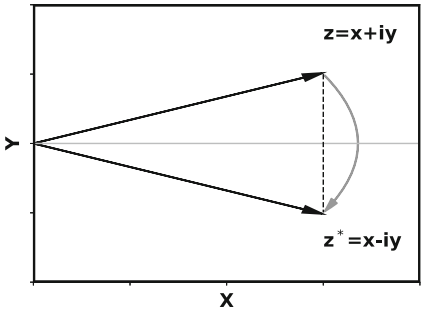


Fig. 5.3 Graphic to illustrate the operation of complex conjugation. Here, we start with a complex number above the horizontal axis and reflect it below when we take the complex conjugate. If we take the complex conjugate again, we return to our original number because a second reflection just reflects back

procedure for the norm produces the result you already know from the conversion between Cartesian and polar coordinates. The complex conjugate in zz^* ensures we get $r^2 = x^2 + y^2$ for the squared radius. For example, $|3 + 4i| = \sqrt{3^2 + 4^2} = 5$.

There will be more to discuss about complex numbers, but we need to know more about exponentials before we can finish this discussion. We will be covering the exponential in the next section. But, before leaving, we have a few more operations to show you. The first is called *rationalizing the denominator* of a complex number. Suppose I have $\frac{1}{z} = \frac{1}{x+iy}$. I can "multiply by one," using the complex conjugate to find

$$\frac{1}{z} \cdot \frac{z^*}{z^*} = \frac{z^*}{|z|^2} = \frac{x - iy}{x^2 + y^2}. \tag{5.11}$$

Note how we got rid of all i's in the denominator. This procedure of "bringing the i up from the denominator" is called rationalizing the denominator. As an example, see if you can verify that $\frac{1}{1-i} = \frac{1}{2}(1+i)$. Remember well the multiply-by-one trick. We will see it return many, many times.

Determining the real and imaginary parts of complex numbers is made easier by using the complex conjugate. For example, you should be able to immediately verify that

$$\operatorname{Re} z = \frac{1}{2}(z + z^*), \quad \text{and} \quad \operatorname{Im} z = \frac{1}{2i}(z - z^*). \tag{5.12}$$

I hope you are becoming comfortable with complex numbers, especially since you have been working with them already as arrows. They may appear a bit forbidding with the i's around, but if you ever feel you are getting lost, remember that they are just those arrows you worked with in Chaps. 1–4.

There is one final form to write down, which is the polar form of a complex number. A little trigonometry shows that

$$z = x + iy = r(\cos\theta + i \sin\theta) \tag{5.13}$$

is the result. In the next section, we will use exponentials of complex numbers to write this another way; if you are familiar with the Euler identity, you will already recognize it above. But first we need to refresh our memory about exponentials, and we need to derive some useful identities. The next section gets a little technical, and you may find it useful to review proof by induction and the binomial theorem before reading it. The exponential function is important, so let us plunge in once you have finished the practice problems below. The video An Introduction To Complex Numbers https://tinyurl.com/r23yn7nw provides additional context for you.

Section 5.2 Problems:

5.2.1 Complex Addition

Compute $(1 - 3i) + (2 + 4i)$.

5.2.2 Complex Multiplication

Compute $(1-3i)(2+4i)$. Do this first in the Cartesian way using $i^2 = -1$ and then do it the polar way. Verify that the two ways give the same answer.

5.2.3 Norm I

Determine $|6 - 8i|$.

5.2.4 Rationalizing the Denominator

Rationalize the denominator of $\frac{1}{2-7i}$.

5.2.5 Norm II

Compute $\left|\frac{1}{1-i}\right|$ by computing $\sqrt{zz^*}$.

5.2.6 Powers of i

Compute i^2, i^3, and i^4. How does $\frac{1}{i}$ compare to i?

5.2.7 Polar to Cartesian

A complex number has $r = 2$ and $\theta = \frac{\pi}{3}$ (the angle is expressed in radians not degrees). Determine the Cartesian form.

5.2.8 Cartesian to Polar

Convert $4 - 3i$ to polar form.

5.2.9 Complex Conjugate

Which complex numbers are equal to their complex conjugate $z = z^*$? Which complex numbers are the opposite of their complex conjugate $z = -z^*$?

5.3 The Exponential Function

The *exponential* function e^x is one of the most important functions in math, and it is used widely in quantum mechanics, so we want to ensure we understand its properties well. When x is real, it describes exponential growth for $x > 0$ and exponential decay for $x < 0$. When it is imaginary, the function oscillates—all of this behavior is important for quantum mechanics. Let us begin by considering raising a number to a positive integer power. For example,

$$\begin{aligned} 2^{x+y} &= \underbrace{2 \times 2 \times 2 \times 2 \cdots \times 2}_{x+y \text{ terms}} \\ &= \underbrace{2 \times 2 \times \cdots \times 2}_{x \text{ terms}} \times \underbrace{2 \times 2 \times \cdots \times 2}_{y \text{ terms}} \\ &= 2^x \times 2^y = 2^x 2^y. \end{aligned} \tag{5.14}$$

While we showed this explicitly for the number 2 raised to any positive integer, the exponent can be zero ($2^0 = 1$) or negative $2^{-x} = \frac{1}{2^x}$). We can also relax the condition that x and y are integers and use the property $2^{x+y} = 2^x 2^y$ for arbitrary real exponents. We will use this fundamental property of exponents to define what e^x is. Namely, we require $e^{x+y} = e^x e^y$ and we require the slope of e^x to be equal to 1 at $x = 0$. This latter condition will determine the value of e. A plot showing how this works is given in Fig. 5.4. Note how the exponential of e^x lies in between 2^x and 4^x. We adjust the value of e until the slope is exactly 1 (corresponding to the dashed line) at $x = 0$. This ends up yielding a transcendental number approximately equal to $2.718281828\cdots$. We will show how to determine this value for e soon.

Our next goal will be to find what is called a power series (or Taylor series) expansion for e^x. If you have taken calculus, you probably learned a general methodology for how to find power series expansions for arbitrary functions. Here, we will use the binomial theorem to determine the power series for the exponential function without calculus.

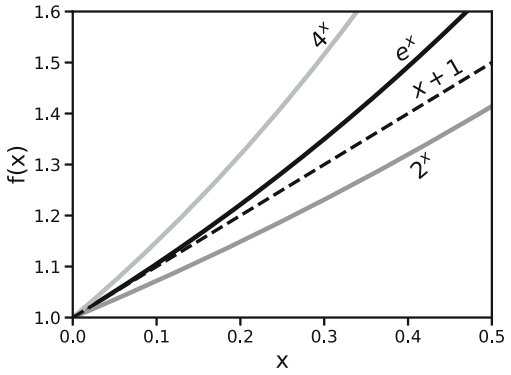


Fig. 5.4 Illustration of how the exponential function is defined. We show 2^x (dark gray), 4^x (light gray), e^x (solid black), and $x + 1$ (dashed black). The function e^x is defined so that its slope at $x = 0$ is 1 (the same slope as the function $x + 1$)

To start, we need to establish the binomial theorem, which is given by

$$(r+s)^n = \sum_{m=0}^{n} \frac{n!}{m!(n-m)!} r^{n-m} s^m, \tag{5.15}$$

where $n! = 1 \times 2 \times 3 \times \cdots \times (n-1) \times n$ is the product of the first n integers and we are using the summation symbol $\sum$, which is a compact way of expressing a summation $\sum_{m=0}^{n} f(m) = f(0) + f(1) + \cdots + f(n)$. Note further that we define $0! = 1$ by convention.

We prove this formula by induction. The way a proof by induction works is to first establish a base case and then assume it holds for all integers from the base case to an integer n and use only those results to prove it also holds for $n + 1$. For example, if the base case corresponds to $n = 0$, then the proof for "$n+1$" establishes it for $n = 1$. Then, the proof for "$n + 1$" establishes it for $n = 2$, and so on, without ever ending. Think about this carefully to see why it works and establishes the proof for all values of n. Bill Courtney gave me a great analogy to think of when proving by induction—consider a line of dominoes all in a row; the base case is like tipping over the first domino and, the proof for $n + 1$ is like we assume the first n dominoes have also fallen; can we show the $n + 1$st will fall as well? if yes, then we know all of the dominoes will fall and the proof is complete. The summation in Eq. (5.15) may seem foreign to you. Whenever we encounter a complex formula, we should examine it for simple cases. For example, when $n = 0$, there is one term in the sum, when $n = 1$, there are two terms, and when $n = 2$, there are three:

$$n = 0 : (r+s)^0 = \sum_{m=0}^{0} \frac{0!}{m!(0-m)!} r^{n-m} s^m = \frac{0!}{0!0!} r^0 s^0 = 1 \tag{5.16}$$

$$n = 1 : (r+s)^1 = \sum_{m=0}^{1} \frac{1!}{m!(1-m)!} r^{n-m} s^m = \frac{1!}{0!1!} r^1 s^0 + \frac{1!}{1!0!} r^0 s^1 = r + s \tag{5.17}$$

$$n = 2 : (r+s)^2 \quad = \sum_{m=0}^{2} \frac{2!}{m!(2-m)!} r^{n-m} s^m = \frac{2!}{0!2!} r^2 s^0 + \frac{2!}{1!1!} r^1 s^1 + \frac{2!}{2!0!} r^0 s^2$$
$$= r^2 + 2rs + s^2. \tag{5.18}$$

Please look carefully at how we go from the summation rules to the explicit terms. Also recall these final results are what you should already know about $(r+s)^n$. You may have even heard of the relationship of these results to Pascal's triangle, but we will not elaborate further on that here.

Now, we go back to our proof by induction. The base case for the binomial theorem corresponds to $n = 0$, which properly expresses $(r+s)^0 = 1$, as we directly evaluated above. Now, we assume the binomial theorem holds for all integers from 0 to n and use that to establish the result for $n+1$. This will then complete the proof.

The proof then proceeds as follows:

$$\begin{aligned}(r+s)^{n+1} &= (r+s)(r+s)^n \\ &= (r+s) \sum_{m=0}^{n} \frac{n!}{m!(n-m)!} r^{n-m} s^m \\ &= \sum_{m=0}^{n} \frac{n!}{m!(n-m)!} r^{n-m+1} s^m + \sum_{m=0}^{n} \frac{n!}{m!(n-m)!} r^{n-m} s^{m+1}, \end{aligned} \tag{5.19}$$

where we used the binomial theorem result for $(r+s)^n$ in the second line and distributed the multiplication in the third line. Next, we use our index shifting to replace $m \to m' - 1$ in the second term to yield

$$(r+s)^{n+1} = \sum_{m=0}^{n} \frac{n!}{m!(n-m)!} r^{n+1-m} s^m + \sum_{m'=1}^{n+1} \frac{n!}{(m'-1)!(n+1-m')!} r^{n+1-m'} s^{m'}. \tag{5.20}$$

Be sure to verify the sum over m' is identical to our previous sum over m for the second term. We now use the "multiply-by-one" trick to rearrange the fractions to look more like the $n+1$ binomial coefficient via

$$\begin{aligned}\frac{n!}{(m'-1)!(n+1-m')!} &= \frac{n!}{(m'-1)!(n+1-m')!} \cdot \frac{n+1}{n+1} \cdot \frac{m'}{m'} \\ &= \frac{(n+1)!}{m'!(n+1-m')!} \cdot \frac{m'}{n+1} \text{ for } 1 \le m' \le n+1, \end{aligned} \tag{5.21}$$

where we used the facts that $n! \cdot (n+1) = (n+1)!$ and $(m'-1)! \cdot m' = m'!$. Note that the summation over m' runs from 1 to $n+1$. Because of the fraction on the right,

we can use the add-zero trick to extend the sum to also include $m' = 0$, because that term in the summation is equal to zero; this results in a summation from 0 to $n + 1$. The second term then becomes

$$\sum_{m'=0}^{n+1} \frac{(n+1)!}{m'!(n+1-m')!} \frac{m'}{n+1} r^{n+1-m'} s^{m'} = \sum_{m=0}^{n+1} \frac{(n+1)!}{m!(n+1-m)!} \frac{m}{n+1} r^{n+1-m} s^{m}, \tag{5.22}$$

because the summation index is a "dummy index" that can be replaced by any symbol we like (here, we choose m for the index). For the first term in Eq. (5.20), we "multiply by one" again (with a different expression for one) to find

$$\frac{n!}{m!(n-m)!} = \frac{n!}{m!(n-m)!} \cdot \frac{n+1}{n+1} \cdot \frac{n+1-m}{n+1-m} = \frac{(n+1)!}{m!(n+1-m)!} \cdot \frac{n+1-m}{n+1}. \tag{5.23}$$

Note that this last step is subtle here. When we multiplied by $(n+1-m)/(n+1-m)$, we restricted to $m \leq n$, so that the denominator is never zero. But, when we look at the term on the right-hand side, because of our convention that $0! = 1$, we can evaluate the expression for $m = n + 1$. We find the result is zero there, so we can add it into the summation, letting the first sum now run to $n + 1$ as well. The final step of the proof is to simply add these two terms in Eqs. (5.22) and (5.23) together with the summation running from $m = 0$ to $m = n + 1$:

$$\begin{aligned}(r+s)^{n+1} &= \sum_{m=0}^{n+1} \frac{(n+1)!}{m!(n+1-m)!} \left(\frac{n+1-m}{n+1} + \frac{m}{n+1} \right) r^{n+1-m} s^{m} \\ &= \sum_{m=0}^{n+1} \frac{(n+1)!}{m!(n+1-m)!} r^{n+1-m} s^{m},\end{aligned} \tag{5.24}$$

which is the result we wanted to show (i.e., the binomial theorem for $n + 1$). This then completes the proof of the binomial theorem. There are a number of steps here, with some of them being a bit subtle, so review this proof carefully to be sure you understand how all these steps fit together. That proof was long! It is one of the pieces we need to establish the power series of the exponential.

We are now ready to verify the power series for the exponential function. We will show that the correct form is given by

$$e^{x} = \sum_{n=0}^{\infty} \frac{1}{n!} x^{n} \tag{5.25}$$

because this power series satisfies the fundamental exponential relation $e^{x+y} = e^{x} \cdot e^{y}$. A sum that has infinity as the upper limit means we continue to sum over all possible values of n without ever ending. The summation must converge to a

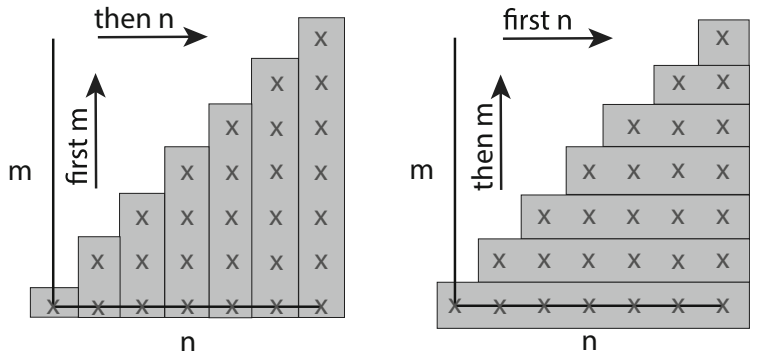


Fig. 5.5 Schematic for how the summations work if m is summed before n (left) or n is summed before m (right). The $\times$ symbols represent the m and n values, while the gray bars indicate how the summation is organized. On the right, we have $0 \leq m \leq n$ and $0 \leq n \leq \infty$, while on the left, we have $m \leq n \leq \infty$ and $0 \leq m \leq \infty$

finite result as we do this in order for such a summation to be meaningful. This will happen with this power series. We start by evaluating the exponential on the left using our power series *ansatz*:

$$e^{x+y} = \sum_{n=0}^{\infty} \frac{1}{n!}(x+y)^n = \sum_{n=0}^{\infty} \frac{1}{n!} \sum_{m=0}^{n} \frac{n!}{m!(n-m)!} x^{n-m} y^m; \tag{5.26}$$

ansatz is a great German word for "educated guess." Note that the two $n!$ terms cancel in the expression on the far right. A graphic depiction of the summation is given in the left panel of Fig. 5.5. The sum over n ranging from 0 to ∞ and over m ranging from 0 to n, with the m sum being performed first, is indicated schematically with the vertical rectangles. The $\times$ marks are the integer m and n values in the summation. For example, pick $n = 0$, then the only value for m is $m = 0$. Next, pick $n = 1$. Then, we have both $m = 0$ and $m = 1$. For $n = 2$, we have $m = 0$, $m = 1$, and $m = 2$. The procedure continues in a similar fashion and is illustrated with the vertical rectangles. We have an infinite number of terms, because we run n from 0 to ∞.

Our next step is straightforward to carry out, but it may seem like it comes from thin air—we switch the order of the summations—summing over n first and then m. When making the switch, we need to ensure that we capture the contribution from every $\times$ mark in the left panel and no more. This is illustrated graphically in the right panel with the horizontal rectangles. For example, if $m = 0$, then $n = 0, 1, 2, \cdots, \infty$, while $m = 1$ gives $n = 1, 2, \cdots, \infty$, and $m = 2$ gives $n = 2, 3, \cdots, \infty$, and so on. It may seem odd that when we sum over m first, we have a finite sum (m runs from 0 to n) followed by an infinite sum (n runs from 0 to ∞), but when we sum over n first, we have two infinite sums—n runs from m to ∞ and m runs from 0 to ∞. One can directly see, by looking at the $\times$'s in the figure

that either way of doing this (m first or n first) *includes exactly the same set of* $\times$*'s*. Hence, the two ways of doing the summation are identical—we can perform the summation either way. Let us convert from an m-first inner summation to an n-first inner summation. We find that we can express e^{x+y} as

$$e^{x+y} = \sum_{m=0}^{\infty} \frac{1}{m!} \sum_{n=m}^{\infty} \frac{1}{(n-m)!} x^{n-m} y^m. \tag{5.27}$$

Note that the way this double summation works is that we start with a particular value of m (which will range from 0 to ∞); then for the given m, we sum n from m to ∞.

We need to massage this formula a bit to get it into the form we need. We will do some index shifting. Let $n' + m = n$, or $n' = n - m$. Then, since $n \geq m$, we have $n' \geq 0$ and

$$e^{x+y} = \underbrace{\sum_{m=0}^{\infty} \frac{1}{m!} y^m}_{e^y} \cdot \underbrace{\sum_{n'=0}^{\infty} \frac{1}{n'!} x^{n'}}_{e^x} = e^x \cdot e^y. \tag{5.28}$$

Note how the first summation is equal to e^y and the second summation is equal to e^x. So, we have shown that this power series satisfies the exponential property $e^{x+y} = e^x \cdot e^y$, which is the first property that the exponential function must satisfy.

To verify the second condition, we need to show the slope of e^x is equal to one at $x = 0$. Recall that the slope of a line is given by the rise divided by the run. When a curve is not a perfect straight line, we find that it often looks closer and closer to a straight line the closer we zoom into a particular point (see Fig. 5.6). Consider the slope for the e^x curve at the points x and $x + \Delta x$, when Δx is small. We find

$$\text{slope} = \frac{\text{rise}}{\text{run}} = \frac{e^{x+\Delta x} - e^x}{(x + \Delta x) - x} = \frac{e^x \left(e^{\Delta x} - 1\right)}{\Delta x}. \tag{5.29}$$

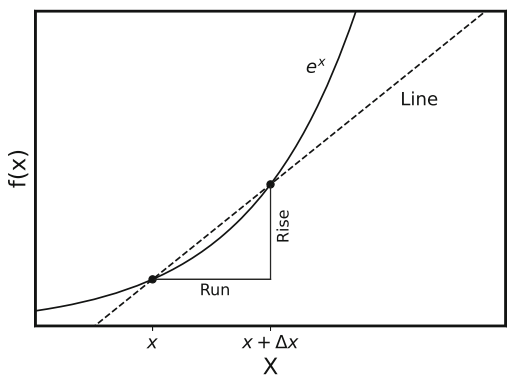


Fig. 5.6 Schematic for how we determine the tangent to a curve at a particular point. The curve e^x is the solid line, while the dashed line is a line that approximates the curve. The line goes through the two function values at x and $x + \Delta x$. By making Δx smaller and smaller, we approach the tangent line. The slope is given by the rise divided by the run

Now, we use the power series we just found to evaluate this result. The second factor in the numerator can be evaluated by noting that

$$e^{\Delta x} - 1 = \sum_{n=0}^{\infty} \frac{1}{n!}(\Delta x)^n - 1 = \cancel{1} + \Delta x + \frac{1}{2}(\Delta x)^2 + \frac{1}{6}(\Delta x)^3 + \cdots - \cancel{1}$$

$$= \Delta x + \frac{1}{2}(\Delta x)^2 + \cdots , \tag{5.30}$$

so the slope at x is equal to $e^x(1 + \frac{1}{2}\Delta x + \frac{1}{6}(\Delta x)^2 + \cdots)$. Now, the next step is similar to what we did when we evaluated the geometric series. We take the limit as $\Delta x \to 0$ to find the tangent of e^x at x (if you like, this is the definition of what the tangent is at x or what the slope is at x). We find

$$\text{slope at } x = e^x. \tag{5.31}$$

I hope you see that the fact that the slope of e^x is equal to e^x is a really cool property of the exponential function. Finally, we set $x = 0$ to find that $e^0 = \sum_{n=0}^{\infty} \frac{1}{n!}0^n = 1$; recall that $0^0 = 1$, $0^n = 0$ for $n \geq 1$, and $0! = 1$. Hence, the slope of e^x at $x = 0$ is 1, as we claimed. This completes the proof that

$$e^x = \sum_{n=0}^{\infty} \frac{1}{n!}x^n. \tag{5.32}$$

Setting $x = 1$ then determines the value for e. We find that

$$e = \sum_{n=0}^{\infty} \frac{1}{n!} = 1 + 1 + \frac{1}{2} + \frac{1}{6} + \frac{1}{24} + \cdots \approx 2.71828\cdots . \tag{5.33}$$

The number e is often defined via

$$e = \lim_{n\to\infty} \left(1 + \frac{1}{n}\right)^n , \tag{5.34}$$

which originally was given by Bernoulli to help solve the compound interest problem. We show this gives the same result for e as the power series does. As $n \to \infty$, $\frac{1}{n}$ is small, and we have already learned that the exponential of a small number can be approximated from the first few terms of the power series. So

$$e^{\frac{1}{n}} \approx 1 + \frac{1}{n} + \frac{1}{2n^2} + \cdots \tag{5.35}$$

gets closer and closer to $1 + \frac{1}{n}$ as n is made larger (because $\frac{1}{2n^2}$ is then very small). Hence

$$\lim_{n\to\infty}\left(1+\frac{1}{n}\right)^n = \lim_{n\to\infty}\left(e^{\frac{1}{n}}\right)^n = \lim_{n\to\infty} e^{\frac{n}{n}} = e^1 = e. \tag{5.36}$$

This example for how to take a limit is quite important. The strategy is that we focus on the *leading-order* behavior and drop all higher-order corrections when the leading correction becomes small. This is the methodology you should use when evaluating limits. Our result can be immediately generalized to an alternative form for the exponential function, namely,

$$e^x = \lim_{n\to\infty}\left(1+\frac{x}{n}\right)^n. \tag{5.37}$$

Keeping in mind the idea of leading-order behavior and higher-order corrections, we can compute the slope of the exponential at $x = 0$ from the power series expression as well. Note that

$$e^x = 1 + x + \frac{x^2}{2} + \cdots, \tag{5.38}$$

and so, for small x, we find that $e^x \approx 1 + x$. This is a linear function with an intercept of one and a slope (given by the coefficient of the x term) of one. So, this is yet another way to see that the slope of e^x at $x = 0$ is one. Seeing things in more than one way is often useful and can even lead to new results. For example, we can use this leading-term method to show that the slope of $e^{ax} = 1 + ax + \cdots$ is equal to a at $x = 0$; using $e^{ax+a\Delta x} = e^{ax}(1 + a\Delta x + \cdots)$ shows the slope is equal to ae^{ax} at arbitrary x.

The slope formula is a very important result. It says that e^x has a slope equal to its value at every point. It is the only function that satisfies this property.

We are now able to identify an important identity called the Euler identity, which describes how e^x behaves for imaginary values of x. It is given by

$$e^{i\theta} = \cos\theta + i\sin\theta, \tag{5.39}$$

for real θ. This is a really amazing identity, involving imaginary numbers, exponentials, and trigonometric functions. We prove it next.

First note that $e^{i\theta}$ is a complex number with a norm (or equivalently a length) of 1. This is because

$$|e^{i\theta}| = \sqrt{z^* z|_{z=e^{i\theta}}} = \sqrt{e^{-i\theta}e^{i\theta}} = \sqrt{e^{-i\theta+i\theta}} = \sqrt{e^0} = 1. \tag{5.40}$$

So, for any real number θ, we have $e^{i\theta}$ lies on the unit circle on the complex plane. When $\theta = 0$, $e^{i0} = 1$, which lies on the horizontal axis. As θ increases, we find $e^{i\Delta\theta} \approx 1 + i\Delta\theta - \frac{1}{2}\Delta^2$ (from the power series formula), which has a positive vertical component, so as θ increases, we move around the unit circle in a counterclockwise direction.

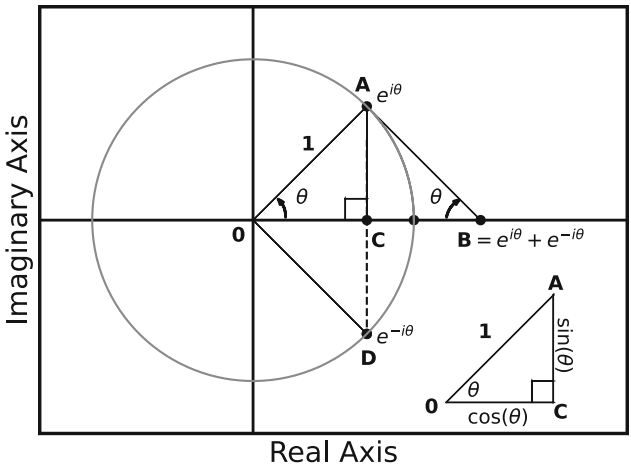


Fig. 5.7 Geometry needed to prove the Euler identity

Now, look at Fig. 5.7. We have drawn $e^{i\theta}$ for a small positive value of θ, located at point A. Then, $e^{-i\theta}$ is the complex conjugate, so it is reflected about the horizontal axis, and then, the sum $e^{i\theta} + e^{-i\theta}$ lies on the horizontal axis (because the imaginary parts cancel in the sum). Note that the triangle OAB is isosceles, since OC=CB (be sure you understand why), and then, the angles θ are the same on the right and on the left. We drop a perpendicular to the horizontal axis from point A, and then, OAC is a right triangle and the line segment OC is half the length of OB. But, OC is defined to be $\cos\theta$, because the hypotenuse of OAC has length 1. Hence, we learn that

$$\cos\theta = \frac{1}{2}(e^{i\theta} + e^{-i\theta}). \tag{5.41}$$

We can go through a similar analysis examining $e^{i\theta} - e^{-i\theta}$ along the imaginary axis and learn that

$$\sin\theta = \frac{1}{2i}(e^{i\theta} - e^{-i\theta}). \tag{5.42}$$

Using these two relations, we solve for $e^{i\theta}$ and find the Euler identity, given in Eq. (5.39). Finally, we find

$$e^{i\theta} = \sum_{n=0}^{\infty} \frac{1}{n!}(i\theta)^n = \underbrace{\sum_{m=0}^{\infty} \frac{1}{(2m)!}(i\theta)^{2m}}_{\text{even } m} + \underbrace{\sum_{m=0}^{\infty} \frac{1}{(2m+1)!}(i\theta)^{2m+1}}_{\text{odd } n}, \tag{5.43}$$

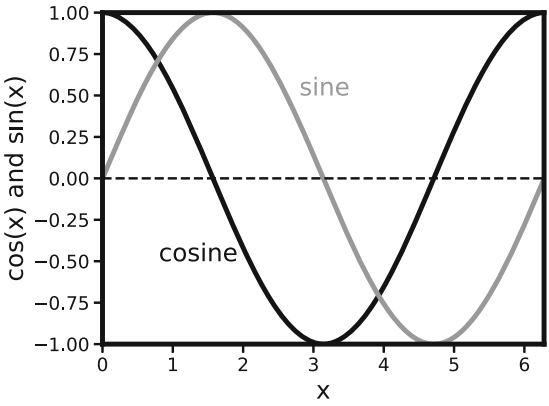


Fig. 5.8 Plots of cosine (black) and sine(gray)

where we separated out the summations over even and odd integers, respectively. Using the fact that $i^2 = -1$, we find the summations have separated into a purely real number and a purely imaginary number, which satisfy

$$\begin{aligned} e^{i\theta} &= \sum_{m=0}^{\infty} \frac{1}{(2m)!}(-1)^m \theta^{2m} + i \sum_{m=0}^{\infty} \frac{1}{(2m+1)!}(-1)^m \theta^{2m+1} \\ &= \cos\theta + i \sin\theta, \end{aligned} \tag{5.44}$$

because the real and imaginary parts are separately equal. Hence, we learn that

$$\cos\theta = \sum_{m=0}^{\infty} \frac{1}{(2m)!}(-1)^m \theta^{2m} = 1 - \frac{1}{2}\theta^2 + \frac{1}{24}\theta^4 + \cdots \tag{5.45}$$

and

$$\sin\theta = \sum_{m=0}^{\infty} \frac{1}{(2m+1)!}(-1)^m \theta^{2m+1} = \theta - \frac{1}{6}\theta^3 + \frac{1}{120}\theta^5 + \cdots . \tag{5.46}$$

Note how $\cos\theta \to 1$ as $\theta \to 0$ and $\sin\theta \to \theta$ as $\theta \to 0$, as shown in Fig. 5.8. The angle θ in this formula is measured in radians, not degrees.

The Euler formula plays a major role in the polar form of complex numbers. We write the Cartesian and polar forms as

$$z = x + iy = re^{i\theta}, \tag{5.47}$$

where $r = \sqrt{zz^*} = \sqrt{|z|^2} = \sqrt{x^2 + y^2}$, $x = r\cos\theta$, $y = r\sin\theta$, and $\tan\theta = \frac{y}{x}$, or $\theta = \tan^{-1}\frac{y}{x}$. In the polar form, we can multiply complex numbers most simply—multiply the magnitude of the arrows and add the times on the clocks. Note that the

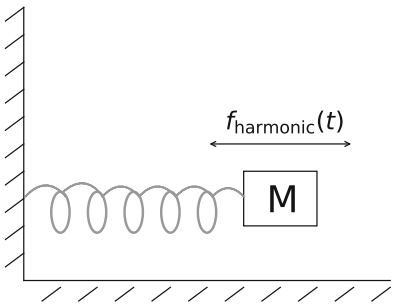


Fig. 5.9 Schematic of harmonic variation with a mass on the end of a spring

polar form for $r = 1$, namely, $e^{i\theta}$ is often called the *complex phase factor* or just *complex phase*, for short.

The final property we discuss in this section is a property called *harmonic variation*. As you have probably seen before, we can represent a harmonically varying function (say, for a mass oscillating on a spring) as a simple cosine function:

$$f_{\text{harmonic}}(t) = A\cos\omega t, \tag{5.48}$$

where A is the amplitude and ω is the angular frequency. We also can write this as

$$f_{\text{harmonic}}(t) = \text{Re}\left(Ae^{-i\omega t}\right); \tag{5.49}$$

here, Re denotes the real part of the complex number, and we require A to be real. The value of f is the position of the center of the mass as it oscillates on the end of the spring with respect to its rest length (see Fig. 5.9).

The mass on the end of the spring oscillates to and fro about the rest length l_0 of the spring. The angular frequency of the oscillation is ω, which is related to the period of the oscillation via $\omega t_{\text{per}} = 2\pi$.

In addition to the position versus time, we also would like to know the velocity versus time. This is given by the rate of change of the position versus time. If you know calculus, this is a simple calculation via taking a derivative with respect to time. But we can also determine it without calculus, because we know the slope of e^x at x. The velocity is given by the slope of the curve of position versus time. Since the power series expansion tells us that $e^{-i\omega t} \approx 1 - i\omega t + \cdots$, we find the slope of a harmonically varying function is given by $-i\omega$ times the function, because the slope of $e^{-i\omega t}$ is $-i\omega e^{-i\omega t}$. Hence, using $x(t) = \text{Re}\left(Ae^{-i\omega t}\right)$, we find that $v(t) = \text{Re}\left(-i\omega e^{-i\omega t}\right) = -A\omega\sin\omega t$; recall that we already showed that the slope of e^{ax} is ae^{ax} at x. This implies that a harmonically varying function (when expressed as an exponential) has a slope given by $-i\omega$ times the function. This property will play a critical role for us in deriving the position-momentum commutation relations in Chap. 7.

One question you might ask is why did we use $e^{-i\omega t}$ instead of $e^{i\omega t}$? It turns out, for this problem, either approach works, but because time advances forward in

one direction, the choice $e^{-i\omega t}$ is preferred because it describes the motion of waves correctly. You can think of this as just a convention we have chosen, but the reasons why we do this are deeper than just a convention.

We have completed our journey into studying properties of the exponential function for now. While there are a number of unique properties that the exponential satisfies, all these properties can be derived from its two defining properties, as we did here. In the next unit, we examine how the corresponding power series can be used to approximate the exponential function, the cosine function, and the sine function.

Section 5.3 Problems:

5.3.1 Binomial Theorem

This problem asks you to verify the binomial theorem concretely by calculating all of the terms in the summation for $n = 5$. Compute

$$(2+3)^5 = \sum_{m=0}^{5} \frac{5!}{m!(5-m)!} 2^{5-m} 3^m = 3125 \tag{5.50}$$

There are six terms you need to compute in the summation.

5.3.2 Binomial Theorem Symmetry

Prove that

$$(r+s)^n = \sum_{m=0}^{n} \frac{n!}{m!(n-m)!} r^{n-m} s^m = \sum_{m'=0}^{n} \frac{n!}{m'!(n-m')!} s^{n-m'} r^{m'} = (s+r)^n \tag{5.51}$$

This shows that the binomial theorem is symmetric in the order of the two powers.

5.3.3 Slope of the Exponential

Use the definition slope = rise over run to compute the slope of $e^{-i\omega t}$ using the points t_0 and $t_0 + \Delta t$. Use the properties of the exponential, not the power series to compute this. Determine what happens to your result as you take the limit $\Delta t \to 0$.

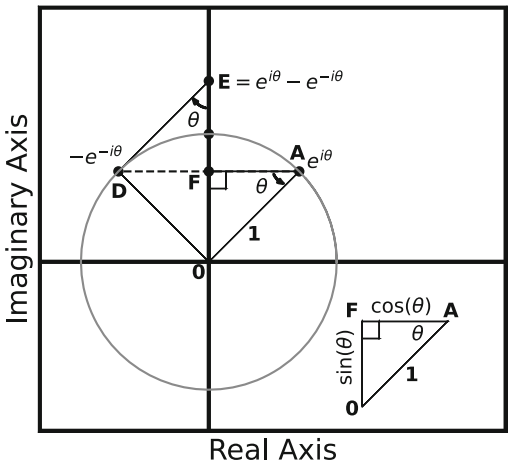


Fig. 5.10 Schematic used for determining the relationship between sine and exponentials

5.3.4 Relationship Between Sine and Exponentials

Using methods similar to those in the main text used to determine the cosine, show that the length of line segment OF is one-half the length of line segment OE and use that to prove that $\sin\theta = \frac{1}{2i}(e^{i\theta} - e^{-i\theta})$. See Fig. 5.10 for the details of the geometry.

5.3.5 Value for e

Using the power series for e^x, compute e to six digits of accuracy, using $x = 1$. Think carefully about how you know when you can stop the calculation because it is accurate enough.

5.3.6 Limits of Powers

Determine $\lim_{n\to\infty}\left(1+\frac{2}{n}\right)^n$.

5.3.7 Trigonometric Identities

Use $\cos\theta = \frac{1}{2}(e^{i\theta} + e^{-i\theta})$ and $\sin\theta = \frac{1}{2i}(e^{i\theta} - e^{-i\theta})$ to verify the following:

- $\cos^2\theta + \sin^2\theta = 1$
- $\cos^2\theta - \sin^2\theta = \cos 2\theta$

- $2\cos\theta\sin\theta = \sin 2\theta$

5.4 Approximations for Exponentials, Sines, and Cosines

We worked hard to establish the power series for the exponential, the sine and the cosine. Now, we want to use them to approximate those functions and learn more about their behavior. We start with the exponential function. In Fig. 5.11, we show plots of e^x, $1+x$, $1+x+\frac{1}{2}x^2$, and $1+x+\frac{1}{2}x^2+\frac{1}{6}x^3$, for real x. These polynomial approximations to the exponential function are found by truncating the power series found in Ch. 5.3 at a particular order of the polynomial. Note that the more terms we keep in the polynomial, the further in x these curves approximate the exponential. For the smallest values of x near zero, the linear approximation suffices, but for larger values, we need more terms.

In many situations, we find we can derive all we need by approximating the exponential by its linear or quadratic approximation only. Recall how we proved the slope of e^x was equal to e^x by computing the rise over the run and taking the limit as $\Delta x \to 0$. We will return again and again to these approximations throughout the book. It is important for you to master them.

The next case we want to look at is $\sin(x)$ and its approximations of x, $x-\frac{1}{6}x^3$, and $x-\frac{1}{6}x^3+\frac{1}{120}x^5$. The comparison is done in Fig. 5.12. Because $\sin(x)$ only has odd powers of x (making it an odd function), the approximations work better than what we saw for the exponential function (Fig. 5.11).

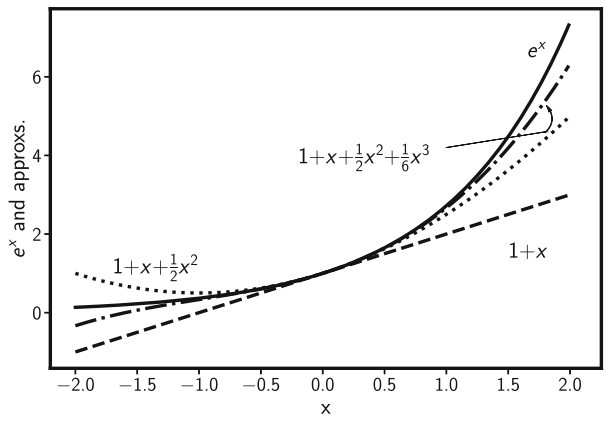


Fig. 5.11 Approximation of the exponential function for real x by truncated polynomials extracted from the power series. The solid line is e^x, the dashed line is $1+x$, the dotted line is $1+x+\frac{1}{2}x^2$, and the chain-dotted line is $1+x+\frac{1}{2}x^2+\frac{1}{6}x^3$

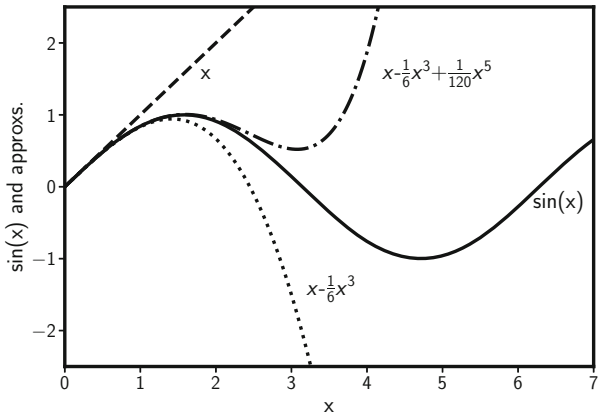


Fig. 5.12 Approximation of $\sin(x)$ for real x by truncated polynomials extracted from the power series. The solid line is $\sin(x)$, the dashed line is x, the dotted line is $x - \frac{1}{6}x^3$, and the chain-dotted line is $x - \frac{1}{6}x^3 + \frac{1}{120}x^5$

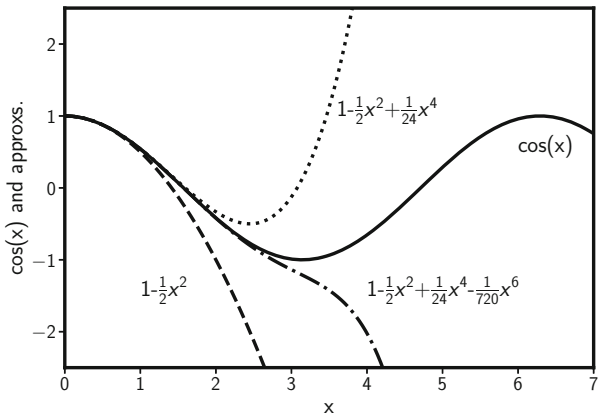


Fig. 5.13 Approximation of $\cos(x)$ for real x by truncated polynomials extracted from the power series. The solid line is $\cos(x)$, the dashed line is $1 - \frac{1}{2}x^2$, the dotted line is $1 - \frac{1}{2}x^2 + \frac{1}{24}x^4$, and the chain-dotted line is $1 - \frac{1}{2}x^2 + \frac{1}{24}x^4 - \frac{1}{720}x^6$

We now move on to $\cos(x)$ and its polynomials of $1 - \frac{1}{2}x^2$ and $1 - \frac{1}{2}x^2 + \frac{1}{24}x^4$ shown in Fig. 5.13. They work well for small x and then fail spectacularly. It looks to me like we are "laying" the approximate polynomial on top of the trigonometric function, and then all of a sudden, it shoots away from it. This spectacular failure, where the polynomial starts to disagree with the trigonometric function, is called a *jet*. Note how the higher-order polynomials for both sine and cosine can start to

produce some of the oscillations of these functions. They produce more and more of these oscillations as we make the degree of the approximating polynomial higher.

This is all we will cover in this short section, but you will want to return to it from time to time to be sure you understand how these functions behave. As you can see, the concepts are straightforward, but it can take some time until you feel comfortable with them and can use them easily. This occurs only with practice. So, let us move on to the problems.

Section 5.4 Problems:

5.4.1 How Accurate?

What polynomial truncation to the exponential function approximates it to an error of at most 1% over the interval $-1 \leq x \leq 1$? Repeat for the interval $-2 \leq x \leq 2$. Note that you should find the error is largest at the end points of the intervals. The percentage error is given by

$$\text{percentage error} = \left| \frac{f_{\text{exact}}(x) - f_{\text{approx}}(x)}{f_{\text{exact}}(x)} \right|. \tag{5.52}$$

You need to verify where the error is the largest and how you know what power to stop the approximation at.

5.4.2 Accuracy for $\sin(x)$

What is the percentage error for the linear, cubic, and quintic approximations to $\sin(x)$ at $x = \frac{\pi}{2}$?

5.4.3 Approximate Trigonometric Identities

One of the important trigonometric identities is $\cos^2 x + \sin^2 x = 1$. If we approximate $\cos x = 1 - \frac{1}{2}x^2$ and $\sin x \approx x$, how large is the error for the trigonometric identity for $0 \leq x \leq \pi$? Repeat this exercise for the higher-order approximations given by $\cos x \approx 1 - \frac{1}{2}x^2 + \frac{1}{24}x^4$ and $\sin x \approx x - \frac{1}{6}x^3$.

5.5 Introduction to 2 × 2 matrices

Matrices are arrays of numbers (they can be restricted to only real numbers or allow complex numbers). For now, we will focus on the simplest of all matrices, the 2 × 2 matrices. In their most general form, the matrix is arranged in an array of four numbers in two rows and two columns:

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \tag{5.53}$$

where a, b, c, and d, are complex numbers. We use subscripts to denote the elements of the matrix. So, A_{ij} is the matrix element in the ith row and jth column. Or, more concretely, we write

$$A = \begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix}, \tag{5.54}$$

with $A_{11} = a$, $A_{12} = b$, $A_{21} = c$, and $A_{22} = d$. We call the matrix elements where $i = j$ the *diagonal elements* (A_{11} and A_{22} or a and d), while the ones with $i \neq j$ are the *off-diagonal elements* (A_{12} and A_{21} or b and c). If $A_{ij} = A_{ji}$ for all i and j, we say we have a *symmetric matrix*; this requires $b = c$ for a 2 × 2 matrix. We can transform a matrix by interchanging the off-diagonal elements (which can be though of as reflecting the elements along the diagonal). This matrix is called the transpose matrix, denote A^T. It takes the form

$$A^T = \begin{pmatrix} a & c \\ b & d \end{pmatrix}. \tag{5.55}$$

You can also think of this as exchanging the rows of the initial matrix with the columns of the transposed matrix. Note that a symmetric matrix satisfies $A = A^T$.

There is a more complex object called the *Hermitian conjugate* of a matrix, which is closely related to the transpose. Indeed, it is formed by transposing the matrix and then taking the complex conjugate of all matrix elements ($A^\dagger_{ij} = A^*_{ji}$). The Hermitian conjugate is denoted with a dagger superscript $A^\dagger$ and it becomes

$$A^\dagger = \begin{pmatrix} A^*_{11} & A^*_{21} \\ A^*_{12} & A^*_{22} \end{pmatrix} = \begin{pmatrix} a^* & c^* \\ b^* & d^* \end{pmatrix}. \tag{5.56}$$

A matrix that is equal to its Hermitian conjugate is said to be a Hermitian matrix, and it satisfies $A = A^\dagger$. While it is true that all real symmetric matrices are Hermitian matrices, Hermitian matrices are not always symmetric matrices (they must be real valued to also be symmetric).

Just like we had rules to add and multiply scalar numbers, and vectors, we have the same with matrices. For matrix addition, we add each element "pointwise"

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} + \begin{pmatrix} a' & b' \\ c' & d' \end{pmatrix} = \begin{pmatrix} a + a' & b + b' \\ c + c' & d + d' \end{pmatrix} \tag{5.57}$$

for the addition of two matrices. Obviously, the sum of two matrices commutes, in that $A + A' = A' + A$. We also can multiply matrices by scalars (similar to how we multiply vectors by scalars)—we simply multiply every matrix element by the scalar.

Before ending this brief introduction, we want to define two invariants of matrices—the determinant and the trace. The determinant of a 2×2 matrix is defined via

$$\det A = A_{11}A_{22} - A_{12}A_{21} = ad - bc. \tag{5.58}$$

The trace is just the sum of the diagonal elements of the matrix:

$$\mathrm{Tr}\, A = A_{11} + A_{22} = a + d. \tag{5.59}$$

We will see later why they are called invariants.

Section 5.5 Problems:

5.5.1 Matrix Addition

Compute the matrix addition:

$$\begin{pmatrix} 1 & 4 \\ -2 & 3 \end{pmatrix} + \begin{pmatrix} 0 & i \\ -i & 2 \end{pmatrix} \tag{5.60}$$

Is your matrix sum a symmetric matrix? Is it a Hermitian matrix?

5.5.2 Determinant and Trace I

Compute the determinant and trace of the following four matrices:

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \text{and} \quad \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}. \tag{5.61}$$

These matrices are called the Pauli spin matrices, and we learn a lot more about them in Chap. 6.

5.5.3 Determinant and Trace II

Compute the determinant and trace of the following matrices:

$$\begin{pmatrix} 2 & 1 \\ 1 & \frac{1}{2} \end{pmatrix}, \quad \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad \text{and} \quad \begin{pmatrix} 1 & -2i \\ 3i & 4 \end{pmatrix}. \tag{5.62}$$

5.6 Matrix Multiplication of 2 × 2 matrices

Now, we will show how we multiply 2×2 matrices. The technique involves a "zipper-like" process with rows and columns of the two matrices, and it is easiest to just illustrate it directly. Consider the product of matrices A and B:

$$AB = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} a' & b' \\ c' & d' \end{pmatrix}. \tag{5.63}$$

The A matrix is the unprimed matrix and the B matrix is the primed one. The product is defined to be

$$AB = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} a' & b' \\ c' & d' \end{pmatrix} = \begin{pmatrix} aa' + bc' & ab' + bd' \\ ca' + dc' & cb' + dd' \end{pmatrix} = C. \tag{5.64}$$

This looks like an awful mess until you see the secret for how it works. Let's start with the C_{11} matrix element (top left) of the product. It is formed by multiplying the first row of A times the first column of B as follows:

$$(a \; b) \cdot \begin{pmatrix} a' \\ c' \end{pmatrix} = aa' + bc'. \tag{5.65}$$

This acts like a zipper—I pair the first elements of the row (a) and the column (a'), multiply them together and add them to the same pair multiplied for the second element of the row (b) and column (c'). You also can think of this as a dot product (if you are familiar with dot products) between a row vector and a column vector (at least when the matrix has all real matrix elements—when they are complex, we continue to multiply rows and columns as described above without taking an additional complex conjugate).

To get the C_{12} element (top right), I multiply the first row times the second column, and in general, C_{ij} is found by multiplying the ith row of A with the jth column of B.

If you have never seen this before, it may seem like a complicated and confusing thing to do, but it will turn out to be quite an interesting thing to do, because using this technique as a way to multiply two matrices allows us to solve many interesting

problems and it turns out to be the most useful way to define matrix multiplication as a generalization to how we multiply scalar numbers. It turns out that Heisenberg rediscovered this result in 1925 when he invented matrix mechanics—he did not even know what a matrix was and called his objects tableau!

So, is there a rational origin for where this comes from? Well, we can think of the matrix as a collection of vectors. For the 2×2 matrix, it is either two row vectors or two column vectors:

$$\begin{pmatrix} \boxed{a \;\; b} \\ \boxed{c \;\; d} \end{pmatrix} \quad \text{or} \quad \begin{pmatrix} \boxed{a} & \boxed{b} \\ \boxed{c} & \boxed{d} \end{pmatrix}. \tag{5.66}$$

Now, we know how to multiply vectors together. The procedure, called an inner (or dot) product, is precisely this "zipper dance," where we take a row on the left and a column on the right, pair them, multiply them, and add them. So, matrix multiplication actually has its origin in the way we multiply vectors together. This is illustrated schematically as follows:

$$\begin{pmatrix} \boxed{a \;\; b} \\ \boxed{c \;\; d} \end{pmatrix} \begin{pmatrix} \boxed{a'} & \boxed{b'} \\ \boxed{c'} & \boxed{d'} \end{pmatrix} = \begin{pmatrix} (a\ b) \cdot \begin{pmatrix} a' \\ c' \end{pmatrix} & (a\ b) \cdot \begin{pmatrix} b' \\ d' \end{pmatrix} \\ (c\ d) \cdot \begin{pmatrix} a' \\ c' \end{pmatrix} & (c\ d) \cdot \begin{pmatrix} b' \\ d' \end{pmatrix} \end{pmatrix} = \begin{pmatrix} aa' + bc' & ab' + bd' \\ ca' + dc' & cb' + dd' \end{pmatrix}. \tag{5.67}$$

We will illustrate a concrete example of this later. Note that when the numbers in the matrix are complex, this product of vectors still can be thought of as an inner product, because in an inner product we complex conjugate a left column vector before multiplying it by a right column vector (technically speaking for two vectors, we take the Hermitian conjugate, converting a column vector to a row vector). The conversion from a column vector to a row vector includes a complex conjugation, so when we have a row vector and take its inner product with a column vector, the complex conjugation has already been done for us, and we need not repeat it.

For scalar multiplication of numbers, there is an identity element—the number 1—which does not change any number when it multiplies it: $a1 = 1a = a$. Since I can multiply matrices together, there must be an analog of the identity. We call this the identity matrix I and since

$$AI = IA = A, \tag{5.68}$$

we immediately find that the identity matrix is

$$I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \tag{5.69}$$

by looking at the matrix multiplication formula. Note that I is unique and is the only matrix that satisfies both $AI = IA = A$ for all matrices A.

Armed with an identity, we define an inverse matrix (when an inverse exists) to satisfy

$$AA^{-1} = I \tag{5.70}$$

(technically speaking, this is a right inverse). We now show that it is also a left inverse by employing the associative property of matrix multiplication. Multiply the right-inverse equation by A on the right to find

$$(AA^{-1})A = IA. \tag{5.71}$$

But, the left-hand side can be rewritten, using the associative property as

$$(AA^{-1})A = A(A^{-1}A). \tag{5.72}$$

Now, the right-hand side of Eq. (5.71) is equal to A, which is also equal to the right-hand side of Eq. (5.72). Since $A = AI$, we also find that

$$(A^{-1}A) = I, \tag{5.73}$$

establishing that we have a left inverse as well.

Let us now construct the inverse for our 2×2 matrix directly. With a little trial and error, we can find that

$$A^{-1} = \begin{pmatrix} d & -b \\ -c & a \end{pmatrix} \frac{1}{ad - bc}, \tag{5.74}$$

when $\det A = ad - bc \neq 0$. It is easy to check this

$$\begin{aligned} AA^{-1} &= \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix} \frac{1}{ad - bc} = \begin{pmatrix} ad - bc & 0 \\ 0 & ad - bc \end{pmatrix} \frac{1}{ad - bc} \\ &= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I, \end{aligned} \tag{5.75}$$

as long as our new friend the determinant is nonzero. This turns out to be a general result for all square matrices—a square matrix is invertible if and only if its determinant is nonzero. If $\det A = 0$, there is no inverse. Note that in the verification that we indeed found the matrix inverse, we used the fact that a scalar λ multiplied by a matrix multiplies all elements of the matrix via

$$\lambda A = \begin{pmatrix} \lambda a & \lambda b \\ \lambda c & \lambda d \end{pmatrix}. \tag{5.76}$$

OK. We have found the matrix identity and matrix inverse (when it exists). We also have used the fact that matrix multiplication is associative $(AB)C = A(BC)$. Is this all we need to know about matrix arithmetic? The answer is no. We have not explored one bizarre, but important, property of matrix multiplication—in general, it is not commutative, meaning we often have $AB \neq BA$; that is, A and B do not commute. Let's check to see if this is true. Recall we already computed

$$AB = \begin{pmatrix} aa' + bc' & ab' + bd' \\ ca' + dc' & cb' + dd' \end{pmatrix}. \tag{5.77}$$

Similarly, one can immediately find that

$$BA = \begin{pmatrix} a'a + b'c & a'b + b'd \\ c'a + d'c & c'b + d'd \end{pmatrix}. \tag{5.78}$$

Now, closely compare the matrix elements. You can see they are not the same. For example, we would need $bc' = b'c$, $ab' + bd' = a'b + b'd$, and $ca' + dc' = c'a + d'c$ for $AB = BA$ to hold, in general. One can see that this will not usually be true. *The order of the factors in matrix multiplication matters.* This is a critical rule you need to know to work in quantum mechanics. This is because noncommutativity also occurs with quantum operators, which we will be exploring soon. *You must get used to working on algebra problems where the order of the factors in terms must be kept track of.* This is one of the challenges you must master on the road to becoming a quantum mechanic. Note, however, that matrix multiplication is distributive. Namely, $A(B + C) = AB + AC$ and $(A + B)C = AC + BC$. One just must ensure that we do not change the matrix product orderings in the distributive law.

Let us go through an example. Let

$$A = \begin{pmatrix} 1 & -1 \\ 2 & 0 \end{pmatrix} \text{ and } B = \begin{pmatrix} 3 & 2 \\ -1 & 1 \end{pmatrix}. \tag{5.79}$$

To multiply out $AB = C$, we need four vector-vector inner products:

$$C_{11}: \quad (1 \ -1) \cdot \begin{pmatrix} 3 \\ -1 \end{pmatrix} = 1 \cdot 3 + (-1) \cdot (-1) = 4$$

$$C_{12}: \quad (1 \ -1) \cdot \begin{pmatrix} 2 \\ 1 \end{pmatrix} = 1 \cdot 2 + (-1) \cdot 1 = 1$$

$$C_{21}: \quad (2 \ 0) \cdot \begin{pmatrix} 3 \\ -1 \end{pmatrix} = 2 \cdot 3 + 0 \cdot (-1) = 6$$

$$C_{22}: \quad (2\ \ 0)\cdot\begin{pmatrix}2\\1\end{pmatrix} = 2\cdot 2 + 0\cdot 1 = 4, \tag{5.80}$$

so

$$C = \begin{pmatrix}4 & 1\\ 6 & 4\end{pmatrix}. \tag{5.81}$$

Now, we repeat for $BA = D$:

$$\begin{aligned}
D_{11}: &\quad (3\ \ 2)\cdot\begin{pmatrix}1\\2\end{pmatrix} = 3\cdot 1 + 2\cdot 2 = 7\\
D_{12}: &\quad (3\ \ 2)\cdot\begin{pmatrix}-1\\0\end{pmatrix} = 3\cdot(-1) + 2\cdot 0 = -3\\
D_{21}: &\quad (-1\ \ 1)\cdot\begin{pmatrix}1\\2\end{pmatrix} = (-1)\cdot 1 + 1\cdot 2 = 1\\
D_{22}: &\quad (-1\ \ 1)\cdot\begin{pmatrix}-1\\0\end{pmatrix} = (-1)\cdot(-1) + 1\cdot 0 = 1,
\end{aligned} \tag{5.82}$$

so

$$D = \begin{pmatrix}7 & -3\\ 1 & 1\end{pmatrix}. \tag{5.83}$$

We clearly have $AB \neq BA$.

If this is the first time you have ever performed matrix multiplication, I recommend that you practice by making some examples yourself or finding them from other sources. You need to be facile with matrix multiplication for the work to come.

There are two special types of matrices we need to discuss. They both require us to apply an operation to the matrices—the Hermitian conjugate, constructed via a complex conjugation and a transpose of the matrix (it is also called the Hermitian adjoint); recall the complex conjugate of $z = x + iy$ is $z^* = x - iy$. For example, we have

$$A = \begin{pmatrix}a & b\\ c & d\end{pmatrix} \rightarrow A^\dagger = \begin{pmatrix}a^* & c^*\\ b^* & d^*\end{pmatrix}, \tag{5.84}$$

which corresponds to interchanging the off-diagonal elements ($b \leftrightarrow c$) and taking the complex conjugate of all matrix elements.

There are two special matrices we can now construct from this "dagger" operation. The first special matrix is called a Hermitian matrix, which satisfies $A^\dagger = A$—in Hermitian matrices, the diagonals are real (be sure you understand

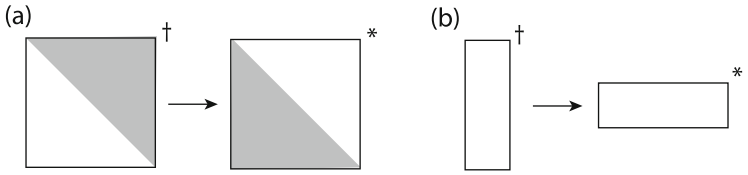


Fig. 5.14 Schematic of Hermitian conjugation. (**a**) Hermitian conjugation of a matrix with transpose and complex conjugation. (**b**) Hermitian conjugation of a vector, which changes from a column to a row, with complex conjugation

why), and the off-diagonals are complex conjugates of each other $A_{ij}^{*} = A_{ji}$ for all $i \neq j$. The second special matrix is called a unitary matrix, and it satisfies $A^{\dagger} = A^{-1}$. These matrices play important roles within quantum mechanics (see Fig. 5.14 to see a schematic for how the Hermitian conjugation works for a matrix). An example of a Hermitian matrix is

$$H = \begin{pmatrix} 1 & i \\ -i & -1 \end{pmatrix} = H^{\dagger}. \tag{5.85}$$

An example of a unitary matrix is

$$U = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ i & -i \end{pmatrix}. \tag{5.86}$$

Unitary and Hermitian properties are not mutually exclusive—there even are matrices we will meet that are both unitary and Hermitian!

Whenever we work with matrices, we need to also be aware of how the matrices relate to vectors and how vectors relate to each other. We end this section with a discussion of the many different products we will encounter between these different objects.

A matrix multiplied times a vector yields a new vector:

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} \alpha \\ \beta \end{pmatrix} = \begin{pmatrix} a\alpha + b\beta \\ c\alpha + d\beta \end{pmatrix}. \tag{5.87}$$

A Hermitian conjugate vector results from taking a column vector, converting it to a row vector via a matrix transpose, and then applying a complex conjugate operation to each element of the row vector (see Fig. 5.14 for a schematic of how this works). So, we have

$$\begin{pmatrix} \alpha \\ \beta \end{pmatrix}^{\dagger} = (\alpha^{*} \quad \beta^{*}). \tag{5.88}$$

The operation also works in the opposite direction. The Hermitian conjugate of a row vector is a column vector, with a complex conjugation, so

$$(\gamma \;\; \delta)^\dagger = \begin{pmatrix} \gamma^* \\ \delta^* \end{pmatrix}. \tag{5.89}$$

The inner product of two vectors is formed similarly to matrix multiplication—we simply think of a column vector as a $1 \times n$ vector, while a row vector is a $n \times 1$ vector. Hence, we first convert the left column vector to the Hermitian conjugate row vector (including complex conjugation), and then, we multiply the row times the column. Be sure to take the complex conjugate before multiplying! We find the inner product between a vector $v = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$ and $v' = \begin{pmatrix} \alpha' \\ \beta' \end{pmatrix}$ satisfies

$$(\alpha^* \;\; \beta^*) \cdot \begin{pmatrix} \alpha' \\ \beta' \end{pmatrix} = \alpha^*\alpha' + \beta^*\beta' = v^\dagger \cdot v'. \tag{5.90}$$

If the inner product of two vectors is equal to 0, then the two vectors are said to be *orthogonal*. The inner product of a vector with itself is always nonnegative—$v^\dagger \cdot v = |\alpha|^2 + |\beta|^2 \geq 0$. This is called the norm squared of the vector (the norm is its square root). The vector-matrix-vector product using a row vector on the left and a column vector on the right becomes a scalar number:

$$\begin{aligned}(\alpha^* \;\; \beta^*) \cdot \begin{pmatrix} a & b \\ c & d \end{pmatrix} \cdot \begin{pmatrix} \alpha' \\ \beta' \end{pmatrix} = (\alpha^* \;\; \beta^*) \cdot \begin{pmatrix} a\alpha' + b\beta' \\ c\alpha' + d\beta' \end{pmatrix} = \alpha^*(a\alpha' + b\beta') \\ + \beta^*(c\alpha' + d\beta') = v^\dagger \cdot A \cdot v'.\end{aligned} \tag{5.91}$$

Finally, an outer product of two vectors (denoted by the $\otimes$ symbol) creates a matrix out of products of the elements of each vector as follows:

$$v \otimes v'^\dagger = \begin{pmatrix} \alpha \\ \beta \end{pmatrix} \otimes (\alpha'^* \;\; \beta'^*) = \begin{pmatrix} \alpha\alpha'^* & \alpha\beta'^* \\ \beta\alpha'^* & \beta\beta'^* \end{pmatrix}. \tag{5.92}$$

We will use these relations throughout the remainder of the book. Note that it takes practice and discipline to get the complex conjugations right.

Section 5.6 Problems:

5.6.1 Is It an Inverse?

Verify that the formula for the inverse works in the form $A^{-1}A = I$ for a general 2×2 matrix that has $ad - bc \neq 0$.

5.6.2 Do They Commute?

Determine whether the following matrices commute (meaning $AB = BA$), by calculating both AB and BA and comparing them:

$$A = \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} \text{ and } B = \begin{pmatrix} 2 & 1 \\ 1 & -2 \end{pmatrix}. \tag{5.93}$$

5.6.3 Find the Inverse

Calculate the inverse of

$$\begin{pmatrix} 1 & -2 \\ -2 & 1 \end{pmatrix} \tag{5.94}$$

and verify that it is indeed the inverse by showing both $AA^{-1} = A^{-1}A = I$.

5.6.4 Is It Unitary?

Let

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}. \tag{5.95}$$

What is the condition on the matrix elements such that A is unitary ($A^\dagger = A^{-1}$)? You will find four conditions. Use the unitary matrix properties to reduce this to two relations between the matrix elements and one relation for the determinant.

5.6.5 Is It Unitary II?

Verify that

$$\frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \text{ and } \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ i & -i \end{pmatrix} \tag{5.96}$$

are unitary matrices (recall how a scalar multiplies a matrix and that $1/i = -i$).

5.6.6 Determinant of a Product Is the Product of the Determinants

Let

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \quad \text{and} \quad B = \begin{pmatrix} a' & b' \\ c' & d' \end{pmatrix}. \tag{5.97}$$

Show that $\det(AB) = \det(A)\det(B) = \det(BA)$.

This is a long calculation. It is best to compute AB first, take its determinant, then compute $\det(A)$ and $\det(B)$, and compute their product. Then, compare both expressions to verify they are the same. Next, compute the product BA, take its determinant, and compare it as well.

This result can be directly generalized to show that the determinant of a product of matrices is equal to the product of the determinants of those matrices. If you are ambitious, you can use the results above in a proof by induction to show that

$$\det(ABC\cdots N) = \det(A)\det(B)\det(C)\cdots\det(N). \tag{5.98}$$

5.6.7 Vector Gymnastics

Let

$$v = \begin{pmatrix} 1 \\ 4 \end{pmatrix}, \quad v' = \begin{pmatrix} 2 \\ -1 \end{pmatrix} \quad \text{and} \quad A = \begin{pmatrix} 0 & -1 \\ -1 & 2 \end{pmatrix}. \tag{5.99}$$

Compute $v^\dagger \cdot v'$, $v \otimes v'^\dagger$, and $v^\dagger \cdot A \cdot v'$.

Hint: $v'^\dagger = (2 \ -1)$.

5.6.8 Orthogonality

Which pairs of vectors are orthogonal

$$v_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad v_w = \begin{pmatrix} 1 \\ -1 \end{pmatrix}, \quad v_3 = \begin{pmatrix} 1 \\ i \end{pmatrix}, \quad \text{and} \quad v_4 = \begin{pmatrix} 1 \\ -i \end{pmatrix}. \tag{5.100}$$

5.6.9 Order of Vectors in the Inner Product

If

$$v = \begin{pmatrix} \alpha \\ \beta \end{pmatrix} \quad \text{and} \quad v' = \begin{pmatrix} \alpha' \\ \beta' \end{pmatrix}. \tag{5.101}$$

Determine how to convert $v^\dagger \cdot v'$ into $v'^\dagger \cdot v$? Consider the elements of v and v' as being complex numbers (which includes the possibility that they are real).

Hint: the relationship involves one mathematical operation.

5.6.10 Factorizing a Unitary

Consider the unitary matrix given by

$$U = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta e^{i\phi} & \cos\theta e^{i\phi} \end{pmatrix}, \tag{5.102}$$

with both θ and ϕ real. Show that U can be factorized in the form

$$U = \begin{pmatrix} 1 & a \\ 0 & 1 \end{pmatrix} \begin{pmatrix} b & 0 \\ 0 & \frac{1}{b} \end{pmatrix} \begin{pmatrix} 1 & 0 \\ c & 1 \end{pmatrix} \tag{5.103}$$

and in the form

$$U = \begin{pmatrix} 1 & 0 \\ \alpha & 1 \end{pmatrix} \begin{pmatrix} \beta & 0 \\ 0 & \frac{1}{\beta} \end{pmatrix} \begin{pmatrix} 1 & \gamma \\ 0 & 1 \end{pmatrix}. \tag{5.104}$$

Determine a, b, and c, as well as α, β, and γ in terms of θ and ϕ.
Hint: multiply out the right-hand side of the matrices and force it to work.

5.7 Eigenvectors and Eigenvalues

In this section, we discuss an interesting property of matrices that is an integral part of quantum mechanics—namely, the determination of eigenvalues and eigenvectors. Some people might say that all of quantum mechanics boils down to analyzing eigenvalues and eigenvectors. Why? Because anything we can measure about a quantum system can be represented by a matrix, and the result of the measurement must be one of the eigenvalues associated with that matrix! In general, the matrix is a Hermitian matrix. But, before we describe the details for how this works, we

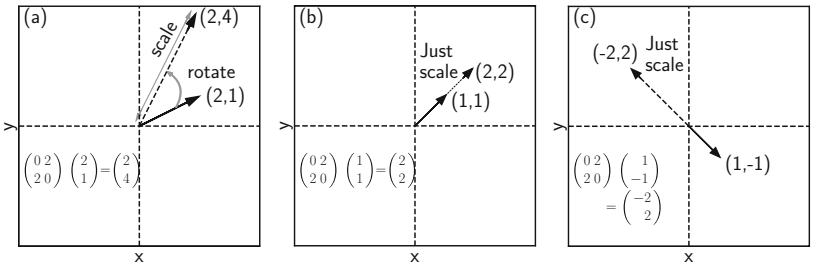


Fig. 5.15 Schematic of how a matrix can rotate and scale a vector after matrix multiplication. In panel (**a**), we show the general case of both scaling and rotation. Panel (**b**) shows a case of just scaling and panel (**c**) also shows just scaling, but here with a negative-valued scaling. The vectors that just scale are called eigenvectors

have to first get through some heavy math to be able to understand this concept. Our approach only uses algebra and the quadratic equation, but it takes time and is long. You have been warned. I do promise that in the end all this heavy lifting will be worth it, but we must get through it first.

The multiplication of matrices times vectors is a physical process—there are two things that can happen to a vector—it can rotate or it can be scaled. In Fig. 5.15, we show works in the general case in panel (a), for a positive scaling in panel (b), and for a negative scaling in panel (c). The case where the matrix just scales a vector with no rotation is a special case we will be discussing in detail in this section. The vectors that just scale are called *eigenvectors*.

In this part of the book, we work primarily with 2×2 (and sometimes 3×3 matrices), but we will focus entirely on 2×2 matrices in this section, which requires us to understand the quadratic equation. We start by deriving the quadratic equation to make sure everyone is familiar with it.

Let $P(x) = ax^2 + bx + c$ be a quadratic polynomial in x, with a, b, and c complex numbers. We want to find the two (possibly complex) solutions to the equation $P(x) = 0$; we denote the solutions as x_+ and x_-. Since $a \neq 0$ (otherwise we have a linear, not a quadratic equation), we divide both sides by a to obtain

$$x^2 + \frac{b}{a}x + \frac{c}{a} = 0. \tag{5.105}$$

The next step is to *complete the square* via our add-zero trick

$$x^2 + \frac{b}{a}x + \frac{c}{a} = \left(x + \frac{b}{2a}\right)^2 + \frac{c}{a} - \frac{b^2}{4a^2} = 0. \tag{5.106}$$

Now, we move the remaining constants to the right-hand side

$$\left(x + \frac{b}{2a}\right)^2 = \frac{b^2 - 4ac}{4a^2} \tag{5.107}$$

and then take the square root of both sides, recalling that there are two square roots—one the negative of the other—yields

$$x + \frac{b}{2a} = \pm\frac{1}{2a}\sqrt{b^2 - 4ac}. \tag{5.108}$$

The two roots are then found by moving the last constant to the right-hand side, which gives us

$$x_\pm = -\frac{b}{2a} \pm \frac{1}{2a}\sqrt{b^2 - 4ac} \tag{5.109}$$

as the two solutions to $P(x) = 0$. Since we can write this equation now as $(x - x_+)(x - x_-) = 0$, we immediately find that $x_+ + x_- = -\frac{b}{a}$ and $x_+ x_- = \frac{c}{a}$.

Now that we reviewed the quadratic equation, we come back to eigenvectors. An *eigenvector* is defined as a vector v associated with a matrix A such that the matrix multiplied by the vector (Av) produces a number (called the *eigenvalue*) times the original vector. While this definition is straightforward, I find this concept is often *very* difficult for students to understand when they first see it. We will go through a number of examples carefully, but it is useful to repeat the definition—*an eigenvector satisfies a matrix times an eigenvector is equal to a number times the original vector*—as many times as you need to to remember it. The strategy we will adopt is to first find the eigenvalue, denoted λ, which satisfies the eigenvalue-eigenvector relationship for the given matrix A, and then, afterward, we will find the eigenvector v. Then, we will have found a solution that solves the relation $Av = \lambda v$.

Recall that multiplying a matrix times a vector is exactly the same as what we did for the first column only of matrix multiplication. Hence

$$\underbrace{\begin{pmatrix} a & b \\ c & d \end{pmatrix}}_{\text{matrix}} \underbrace{\begin{pmatrix} \alpha \\ \beta \end{pmatrix}}_{\text{vector}} = \begin{pmatrix} a\alpha + b\beta \\ c\alpha + d\beta \end{pmatrix} = \underbrace{\lambda}_{\text{number}} \underbrace{\begin{pmatrix} \alpha \\ \beta \end{pmatrix}}_{\text{original vector}} = \begin{pmatrix} \lambda\alpha \\ \lambda\beta \end{pmatrix} \tag{5.110}$$

for an eigenvector $\begin{pmatrix} \alpha \\ \beta \end{pmatrix}$, with λ called the eigenvalue. This appears to be quite hard to solve, since we need both $a\alpha + b\beta = \lambda\alpha$ and $c\alpha + d\beta = \lambda\beta$ to hold simultaneously. So, we proceed carefully.

First notice that if $\alpha = \beta = 0$, the equations we are trying to solve are automatically satisfied for any λ. This is a trivial solution that we are not interested in. To find the nontrivial solution, we use the multiply-by-one trick, in the form $Iv = v$ for the 2×2 identity matrix I, to rewrite the eigenvalue-eigenvector relation as

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} \alpha \\ \beta \end{pmatrix} = \lambda \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} \alpha \\ \beta \end{pmatrix}. \tag{5.111}$$

We next, multiply the identity matrix by λ and subtract the right-hand side from both sides of the equation to yield

$$\begin{pmatrix} a-\lambda & b \\ c & d-\lambda \end{pmatrix} \begin{pmatrix} \alpha \\ \beta \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}. \tag{5.112}$$

We still have the trivial solution for all λ. We want to find the nontrivial solution, in order to determine the eigenvector. If the matrix on the left-hand side was invertible (meaning it has an inverse), then we could multiply both sides of the equation by the inverse matrix, which would yield the trivial solution $\begin{pmatrix} \alpha \\ \beta \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$. The nontrivial solution for the eigenvector occurs *only when the determinant is zero*, implying that the matrix is not invertible. Hence, we require

$$\det \begin{pmatrix} a-\lambda & b \\ c & d-\lambda \end{pmatrix} = (a-\lambda)(d-\lambda) - bc = 0, \tag{5.113}$$

or

$$\lambda^2 - (a+d)\lambda + ad - bc = 0 \tag{5.114}$$

(note that it is not a coincidence that $a+d = \mathrm{Tr}A$ and $ad-bc = \det A$ both appear in this equation). This is solved by the quadratic equation giving

$$\lambda_\pm = \frac{a+d}{2} \pm \frac{1}{2}\sqrt{(a+d)^2 - 4ad + 4bc}. \tag{5.115}$$

Using $(a+d)^2 - 4ad = a^2 + 2ad + d^2 - 4ad = a^2 - 2ad + d^2 = (a-d)^2$ yields

$$\lambda_\pm = \frac{a+d}{2} \pm \frac{1}{2}\sqrt{(a-d)^2 + 4bc}. \tag{5.116}$$

Note that if A is Hermitian (meaning a and d are real and $c = b^*$), then the eigenvalue is always real, because $(a-d)^2 + 4bc = (a-d)^2 + 4|b|^2$ is a real nonegative number. This turns out to be a general property of Hermitian matrices of any dimension.

Let us examine four concrete examples. The first is

$$A = \begin{pmatrix} 0 & 1 \\ -3 & 1 \end{pmatrix}. \tag{5.117}$$

It is not a Hermitian matrix. The eigenvalues are

$$\lambda_{\pm} = \frac{1}{2} \pm \frac{1}{2}\sqrt{1-12} = \frac{1}{2} \pm i\frac{\sqrt{11}}{2}, \tag{5.118}$$

which are complex. The second example is

$$A = \begin{pmatrix} 1 & 2 \\ 2 & -1 \end{pmatrix}, \tag{5.119}$$

which is Hermitian and satisfies

$$\lambda_{\pm} = \pm\frac{1}{2}\sqrt{4+16} = \pm\sqrt{5}. \tag{5.120}$$

This has real eigenvalues. Note that there are two roots, so we expect there to be two eigenvectors associated with the two eigenvalues for this matrix. The third example is a complex-valued matrix, which is Hermitian

$$A \begin{pmatrix} 1 & i \\ -i & -1 \end{pmatrix}; \tag{5.121}$$

its eigenvalues are $\pm\sqrt{2}$. Our final example is a non-Hermitian matrix:

$$A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \tag{5.122}$$

it has one eigenvalue equal to 0. This occurs for the vector $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$—all other vectors of the form $\begin{pmatrix} a \\ b \end{pmatrix}$, with $b \neq 0$, will rotate and scale (pure rotation only occurs if $a = 0$).

Henceforth, we assume A is Hermitian, meaning both a and d are real and $b = c^*$. Then, we find the eigenvalues are real and given by

$$\lambda_{\pm} = \frac{a+d}{2} \pm \frac{1}{2}\sqrt{(a-d)^2 + 4|b|^2}. \tag{5.123}$$

A particularly simple case occurs when $b = 0$. Then, the eigenvalues are equal to a and d and the corresponding eigenvectors can be read off as $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$, because the matrix is diagonal. Please check these by matrix-vector multiplication with $A = \begin{pmatrix} a & 0 \\ 0 & d \end{pmatrix}$.

How do we find the eigenvectors for the general case, when $b \neq 0$? We go back to the starting point in Eq. (5.112) and solve for α and β after plugging in what the

eigenvalue λ is

$$\begin{pmatrix} a-\lambda_\pm & b \\ b^* & d-\lambda_\pm \end{pmatrix} \begin{pmatrix} \alpha \\ \beta \end{pmatrix} = \begin{pmatrix} (a-\lambda_\pm)\alpha + b\beta \\ b^*\alpha + (d-\lambda_\pm)\beta \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}. \tag{5.124}$$

Looking at the top row, we find a relation between the coefficients as

$$\beta = -\frac{a-\lambda_\pm}{b}\alpha. \tag{5.125}$$

Plugging this into the bottom row yields

$$b^*\alpha - \frac{(a-\lambda_\pm)(d-\lambda_\pm)}{b}\alpha = 0. \tag{5.126}$$

This only has a solution for $\alpha \neq 0$ if

$$\frac{|b|^2 - (a-\lambda_\pm)(d-\lambda_\pm)}{b}\alpha = 0. \tag{5.127}$$

But the term in the numerator is always equal to zero, because it was the original quadratic equation condition for the eigenvalues. So, the equation holds.

Something is odd though. We have not found both α and β, just a relationship between the two. This is a general result and always occurs. It is because an eigenvector is not uniquely determined—if I multiply the eigenvector by any nonzero number, it is still an eigenvector. This occurs because the eigenvalue-eigenvector relationship $Av = \lambda v$ is a homogenenous equation. If I multiply both sides by a scalar c, I find $A(cv) = \lambda(cv)$, so that cv is also an eigenevector with the same eigenvalue. This extra freedom can easily lead to confusion, so we have developed a convention—we normalize the eigenvectors. This means we multiply them by a constant so that the norm of the vector is one, or $|\alpha|^2 + |\beta|^2 = 1$. The method used to solve this is simple—just divide by the length. So, for example, if the vector was equal to $(4 \quad -3)$, then the norm is $\sqrt{4^2 + (-3)^2} = \sqrt{25} = 5$. Hence, our normalized vector is $(\frac{4}{5} \quad -\frac{3}{5})$. So, we need to do the same thing for our general eigenvector. We find

$$|\alpha|^2 + |\beta|^2 = |\alpha|^2 + \underbrace{\left|\frac{a-\lambda_\pm}{b}\right|^2 |\alpha|^2}_{|\beta|^2} = 1. \tag{5.128}$$

Rearranging this equation gives

$$|\alpha|^2 \left(1 + \left|\frac{a-\lambda_\pm}{b}\right|^2\right) = |\alpha|^2 \frac{|b|^2 + |a-\lambda_\pm|^2}{|b|^2} = 1. \tag{5.129}$$

Now, we solve by taking a square root and then using the solution for α to find β. This gives us

$$\alpha = \frac{|b|e^{i\theta}}{\sqrt{|b|^2 + |a - \lambda_\pm|^2}} \text{ and } \beta = -\frac{(a - \lambda_\pm)|b|e^{i\theta}}{b\sqrt{|b|^2 + |a - \lambda_\pm|^2}}. \tag{5.130}$$

You might be surprised by the appearance of the $e^{i\theta}$ factor in these expressions. This is called a complex phase, and the normalized eigenvectors are still determined only up to a complex phase (i.e., a complex number $e^{i\theta}$) that multiplies both α and β. It is conventional to pick $\theta = 0$ so that $e^{i\theta} = 1$, in order to make our life easier; this means α is real and β is possibly complex. Once we use this convention, the eigenvectors are completely determined.

We discuss this a bit more, because it is a subtle point, which is often difficult to understand when you first see it. Suppose $\begin{pmatrix} \alpha \\ \beta \end{pmatrix}$ is an eigenvector of our general Hermitian matrix:

$$\begin{pmatrix} a & b \\ b^* & d \end{pmatrix} \begin{pmatrix} \alpha \\ \beta \end{pmatrix} = \lambda \begin{pmatrix} \alpha \\ \beta \end{pmatrix}. \tag{5.131}$$

Then, if we consider a new vector $\begin{pmatrix} \gamma e^{i\theta}\alpha \\ \gamma e^{i\theta}\beta \end{pmatrix}$, with γ and θ real, we find that

$$\begin{pmatrix} a & b \\ b^* & d \end{pmatrix} \underbrace{\gamma e^{i\theta} \begin{pmatrix} \alpha \\ \beta \end{pmatrix}}_{\text{eigenvector}} = \lambda \underbrace{\gamma e^{i\theta} \begin{pmatrix} \alpha \\ \beta \end{pmatrix}}_{\text{eigenvector}} \tag{5.132}$$

is also an eigenvector with the same eigenvalue. This implies that the eigenvector-eigenvalue relationship only determines the ratio of α/β. To get control over these extra degrees of freedom, we require the eigenvector to be unit norm. Hence, if $|\alpha|^2 + |\beta|^2 = 1$, then we must have $\gamma = 1$ to ensure that $\gamma^2(|\alpha|^2 + |\beta|^2) = 1$. Hence, we still have the uncertainty of the complex phase $e^{i\theta}$. The official math language for this is that the eigenvector is a "ray in the Hilbert space." If you are familiar with the concept of an equivalence relation, you can think of the ray as the equivalence class of unit norm vectors—we only need one representative from the equivalence class to represent it, so the vectors related by a complex phase are all "essentially the same."

What this implies is that eigenvectors such as $\frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ 1 \end{pmatrix}$ and $-\frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ 1 \end{pmatrix}$ and $\frac{i}{\sqrt{2}}\begin{pmatrix} 1 \\ 1 \end{pmatrix}$ and $\frac{1+i}{2}\begin{pmatrix} 1 \\ 1 \end{pmatrix}$ and all other choices of the form $\frac{e^{i\theta}}{\sqrt{2}}\begin{pmatrix} 1 \\ 1 \end{pmatrix}$ all refer to the same ray or as we will say in quantum mechanics to the same "state." In other words, two quantum states are the same if they are related by a complex phase $e^{i\theta}$. Similarly, we

say two eigenvectors are the same if $v_1 = \begin{pmatrix}\alpha\\ \beta\end{pmatrix}$ and $v_2 = e^{i\theta}\begin{pmatrix}\alpha\\ \beta\end{pmatrix}$, and then, both v_1 and v_2 *represent the same eigenvector*! The general formulas for the eigenvectors are complicated. The idea is not to memorize them, as it is often easier to rederive them for any specific example that needs to be worked out.

This was a lot of work, but we now need to verify that it is correct. This will require even more work! So, take a deep breath as we check our calculation. We start by multiplying the matrix A times our eigenvector, expressed in the "polite way" (norm equal to one, first nonzero element positive). Using Eq. (5.130), we find

$$\begin{aligned}\begin{pmatrix} a & b\\ b^* & d\end{pmatrix}\begin{pmatrix}\alpha\\ \beta\end{pmatrix} &= \begin{pmatrix} a & b\\ b^* & d\end{pmatrix}\frac{|b|}{\sqrt{|b|^2+|a-\lambda_\pm|^2}}\begin{pmatrix}1\\ -\frac{a-\lambda_\pm}{b}\end{pmatrix}\\ &= \frac{|b|}{\sqrt{|b|^2+|a-\lambda_\pm|^2}}\begin{pmatrix} a-(a-\lambda_\pm)\\ b^*-\frac{d}{b}(a-\lambda_\pm)\end{pmatrix}. \end{aligned} \tag{5.133}$$

Yes, this looks like a mess. What is worse, we need to verify the second line is equal to $\lambda\begin{pmatrix}\alpha\\ \beta\end{pmatrix}$. Fortunately, we can ignore the fraction $|b|/\sqrt{|b|^2+|a-\lambda_\pm|^2}$, when we verify the eigenvalue-eigenvector relationship, because eigenvectors can be multiplied by any nonzero number, and they still remain eigenvectors. Hence, the two equations that must hold are

$$a-(a-\lambda_\pm) = \lambda_\pm \tag{5.134}$$

$$b^* - \frac{d}{b}(a-\lambda_\pm) = -\frac{\lambda_\pm}{b}(a-\lambda_\pm). \tag{5.135}$$

The first equation obviously holds, because $a - a = 0$. As for the second, we multiply both sides by $b \neq 0$, so that the left-hand side becomes

$$\begin{aligned}|b|^2 - d(a-\lambda_\pm) &= |b|^2 - ad + \frac{ad+d^2}{2} \pm \frac{d}{2}\sqrt{(a-d)^2+4|b|^2}\\ &= |b|^2 - \frac{ad}{2} + \frac{d^2}{2} \pm \frac{d}{2}\sqrt{(a-d)^2+4|b|^2}. \end{aligned} \tag{5.136}$$

Now, we simplify the right-hand side of Eq. (5.135). After multiplying by b, it is

$$\begin{aligned}&-\left(\frac{a+d}{2} \pm \frac{1}{2}\sqrt{(a-d)^2+4|b|^2}\right)\left(\frac{a-d}{2} \mp \frac{1}{2}\sqrt{(a-d)^2+4|b|^2}\right)\\ &\quad= -\frac{a^2-d^2}{4} \pm \frac{d}{2}\sqrt{(a-d)^2+4|b|^2} + \frac{1}{4}\left((a-d)^2+4|b|^2\right)\\ &\quad= -\frac{ad}{2} + \frac{d^2}{2} + |b|^2 \pm \frac{d}{2}\sqrt{(a-d)^2+4|b|^2}, \end{aligned} \tag{5.137}$$

which is the same. This means we have verified the eigenvalue-eigenvector relationship.

So, we have shown that these two eigenvectors are

$$e_+ = \frac{|b|}{\sqrt{|b|^2 + (a-\lambda_+)^2}} \begin{pmatrix} 1 \\ -\frac{a-\lambda_+}{b} \end{pmatrix}$$
$$e_- = \frac{|b|}{\sqrt{|b|^2 + (a-\lambda_-)^2}} \begin{pmatrix} 1 \\ -\frac{a-\lambda_-}{b} \end{pmatrix}. \tag{5.138}$$

It turns out that the eigenvectors of Hermitian matrices, corresponding to different eigenvalues, are always orthogonal. This means that $e_+^\dagger \cdot e_- = 0$—recall the $\dagger$ denotes the Hermitian conjugate, which takes the vector $\begin{pmatrix} \alpha \\ \beta \end{pmatrix}$ to its transpose conjugate $(\alpha^* \ \ \beta^*)$. We check the orthogonality next, but we suppress the common factor $|b|^2/\sqrt{(|b|^2 + (a-\lambda_+)^2)(|b|^2 + (a-\lambda_-)^2)}$ in our manipulations. This gives

$$e_+^\dagger \cdot e_- \propto 1 + \frac{(a-\lambda_+)(a-\lambda_-)}{|b|^2} \propto |b|^2 + a^2 - a(\lambda_+ + \lambda_-) + \lambda_+\lambda_-, \tag{5.139}$$

where $\propto$ mean "proportional to" and we removed the common factor of $1/|b|^2$ from the second line. But the eigenvalues of 2×2 matrices satisfy special properties—their sum is $\mathrm{Tr}\, A$ and their product is $\det A$. This is most easily seen by looking at Eq. (5.114), which shows the linear coefficient is $-\mathrm{Tr}\, A$ and the constant term is $\det A$. We can also write that quadratic as $(\lambda - \lambda_+)(\lambda - \lambda_-)$. Expanding the quadratic out and equating the coefficients then tells us that $\lambda_+ + \lambda_- = a + d$ and $\lambda_+\lambda_- = ad - |b|^2$. Using these results in the inner product of the eigenvectors then tells us it is equal to $|b|^2 + a^2 - a(a+d) + ad - |b|^2 = 0$. Hence, the two eigenvectors are indeed orthogonal.

Note that we need to take the Hermitian conjugate of the vector e_+ before performing the inner product, because that is the requirement for the inner product of vectors that have components that are complex valued. The Hermitian conjugate ensures that the norm squared of a vector is positive, as it must be. The fact that eigenvectors form an orthonormal set (they have norm equal to one and are mutually orthogonal) is an important result from linear algebra, which we won't prove in general here, but you can see that it comes out naturally when we examine 2×2 matrices.

Again, that was a lot of work! But we got through it. We will work on building up stamina in algebraic manipulations as we work through this book. Eventually, you will be comfortable working out very lengthy calculations. We have one final topic we need to show about eigenvectors and that is that the set of eigenvectors is also *complete*. This means that the set of eigenvectors spans the vector space. Or, without the jargon, that any vector can be written as a sum of numbers multiplied times the different eigenvectors. When such a construction exists for all vectors in

the vector space, we say we have a basis for the vector space. Here, we will show completeness using a different approach that involves the outer product. One way to state what completeness is, which is the way we will use it in quantum mechanics, is that

$$\mathbb{I}_2 = \begin{pmatrix}\alpha_+\\ \beta_+\end{pmatrix} \otimes (\alpha_+^* \;\; \beta_+^*) + \begin{pmatrix}\alpha_-\\ \beta_-\end{pmatrix} \otimes (\alpha_-^* \;\; \beta_-^*). \tag{5.140}$$

Don't freak out! I will explain what these funny symbols mean and how we verify this identity. The identity matrix $\mathbb{I}_2$ is just the 2×2 identity matrix, given by $\begin{pmatrix}1 & 0\\ 0 & 1\end{pmatrix}$, while the outer product satisfies

$$\begin{pmatrix}\alpha\\ \beta\end{pmatrix} \otimes (\alpha^* \;\; \beta^*) = \begin{pmatrix}\alpha\alpha^* & \alpha\beta^*\\ \beta\alpha^* & \beta\beta^*\end{pmatrix}. \tag{5.141}$$

You can think of the outer product as being the first element of the first column vector (α) multiplied by the second row vector, forming the first row of the outer product, and similarly, the second element of the column vector (β) is multiplied by the second row vector, forming the lower row of the outer product. Schematically, this looks like

$$\begin{pmatrix}\alpha\times\boxed{\alpha^* \;\; \beta^*}\\ \beta\times\boxed{\alpha^* \;\; \beta^*}\end{pmatrix}. \tag{5.142}$$

All outer products (also called tensor products) are formed this way.

After recalling that we are adding two matrices together, the 1 1 element of the sum on the right-hand side can be immediately seen to be

$$|\alpha_+|^2 + |\alpha_-|^2 = \frac{|b|^2}{|b|^2 + (a-\lambda_+)^2} + \frac{|b|^2}{|b|^2 + (a-\lambda_-)^2} = 1 \tag{5.143}$$

$$\frac{|b|^2\left(2|b|^2 + 2a^2 - 2a\,(\lambda_+ + \lambda_-) + \lambda_+^2 + \lambda_-^2\right)}{|b|^4 + |b|^2\left(2a^2 - 2a(\lambda_+ + \lambda_-) + \lambda_+^2 + \lambda_-^2\right) + \left(a^2 - a(\lambda_+ + \lambda_-) + \lambda_+\lambda_-\right)^2} = 1$$

$$\frac{|b|^2\left(2|b|^2 + 2a^2 - 2a(a+d) + a^2 + d^2 + 2|b|\right)^2}{|b|^4 + |b|^2\left(2a^2 - 2a(a+d) + a^2 + d^2 + 2|b|^2\right) + \left(a^2 - a(a+d) + ad - |b|^2\right)^2}$$

$$= \frac{4|b|^4 + |b|^2(a-d)^2}{4|b|^4 + |b|^2(a-d)^2} = 1. \tag{5.144}$$

In these algebraic manipulations, we used the facts that $\lambda_+ + \lambda_- = a + d$, $\lambda_+\lambda_- = ad - |b|^2$, and $\lambda_+^2 + \lambda_-^2 = (a+d)(\lambda_+ + \lambda_-) - 2(ad - |b|^2) = a^2 + d^2 + 2|b|^2$.

Similarly, the 1 2 element of the sum of the two matrices is

$$\alpha_+\beta_+^* + \alpha_-\beta_-^* = |b|^2\left(-\frac{\frac{a-\lambda_+}{b^*}}{|b|^2+(a-\lambda_+)^2} - \frac{\frac{a-\lambda_-}{b^*}}{|b|^2+(a-\lambda_-)^2}\right) \tag{5.145}$$

$$= -\frac{|b|^2}{b^*\left(|b|^2+(a-\lambda_+)^2\right)\left(|b|^2+(a-\lambda_-)^2\right)} \times \Big((a-\lambda_+)\big[|b|^2+(a-\lambda_-)^2\big] + (a-\lambda_-)\big[|b|^2+(a-\lambda_+)^2\big]\Big). \tag{5.146}$$

Next, we remove the common factors and substitute in for $\lambda_\pm$ in the squared terms, which gives

$$\begin{aligned}
&\propto (a-\lambda_+)\left[|b|^2+\left(\frac{a-d}{2}+\frac{1}{2}\sqrt{(a-d)^2+4|b|^2}\right)^2\right] \\
&\quad + (a-\lambda_-)\left[|b|^2+\left(\frac{a-d}{2}-\frac{1}{2}\sqrt{(a-d)^2+4|b|^2}\right)^2\right] \\
&\propto (a-\lambda_+)\left[|b|^2+\frac{(a-d)^2}{4}+\frac{a-d}{2}\sqrt{(a-d)^2+4|b|^2}+\frac{(a-d)^2}{4}+|b|^2\right] \\
&\quad + (a-\lambda_-)\left[|b|^2+\frac{(a-d)^2}{4}-\frac{a-d}{2}\sqrt{(a-d)^2+4|b|^2}+\frac{(a-d)^2}{4}+|b|^2\right]
\end{aligned} \tag{5.147}$$

Now, we substitute in for the remaining $\lambda_\pm$ terms

$$\begin{aligned}
&\propto \left[\frac{a-d}{2}-\frac{1}{2}\sqrt{(a-d)^2+4|b|^2}\right]\left[2|b|^2+\frac{(a-d)^2}{2}+\frac{a-d}{2}\sqrt{(a-d)^2+4|b|^2}\right] \\
&+ \left[\frac{a-d}{2}+\frac{1}{2}\sqrt{(a-d)^2+4|b|^2}\right]\left[2|b|^2+\frac{(a-d)^2}{2}-\frac{a-d}{2}\sqrt{(a-d)^2+4|b|^2}\right].
\end{aligned} \tag{5.148}$$

When we FOIL these terms, we see that the terms proportional to the square root cancel, leaving us with

$$\begin{aligned}
&\propto (a-d)|b|^2+\frac{(a-d)^3}{4}-\frac{(a-d)^3}{4}-(a-d)|b|^2 \\
&\quad + (a-d)|b|^2+\frac{(a-d)^3}{4}-\frac{(a-d)^3}{4}-(a-d)|b|^2 \\
&\propto 0.
\end{aligned} \tag{5.149}$$

Similarly, one can show that the 2 1 element is zero and the 2 2 element is 1. The former follows since it is the complex conjugate of the 1 2 element, while the latter is a problem for you. While the fact that the sum of the outer product of eigenvectors equals the unit matrix may seem like some novelty at this point, it is quite important, and we will use it heavily in the remainder of the book.

OK. Time to relax. We are done with the long algebra for a while. We now examine how these ideas work with an explicit example. Note that once we set specific values for a, b, and d, we could substitute into our general equations and get all the results, but it often is easier to rederive the results for a particular case. So, as an example, let us choose

$$A = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}. \tag{5.150}$$

The eigenvalues $\lambda_\pm$ satisfy $\det \begin{pmatrix} -\lambda & 1 \\ 1 & -\lambda \end{pmatrix} = \lambda^2 - 1 = 0$, so that $\lambda_\pm = \pm 1$. The eigenvectors are found by solving

$$(A - \lambda_\pm \mathbb{I}_2) \begin{pmatrix} \alpha_\pm \\ \beta_\pm \end{pmatrix} = 0 \;\rightarrow\; \begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} \alpha_+ \\ \beta_+ \end{pmatrix} = 0 \text{ and } \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} \alpha_- \\ \beta_- \end{pmatrix} = 0. \tag{5.151}$$

So, we find the eigenvectors to be

$$e_+ = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \text{ and } e_- = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}. \tag{5.152}$$

You can check these results, because $-\alpha_+ + \beta_+ = 0$ and $|\alpha_+|^2 + |\beta_+|^2 = 1$ and $\alpha_- + \beta_- = 0$ and $|\alpha_-|^2 + |\beta_-|^2 = 1$.

We can check that the eigenvectors form an orthonormal set, since $e_+ \cdot e_+ = 1$, $e_- \cdot e_- = 1$ and $e_+ \cdot e_- = 0$ hold. For completeness, we see that

$$e_+ \otimes e_+^\dagger + e_- \otimes e_-^\dagger = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} + \frac{1}{2} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}. \tag{5.153}$$

So, the completeness relation also holds. Since the eigenvectors form a complete set, any two-dimensional vector can be expanded as $v = re_+ + se_-$, with r and s complex numbers. In addition, the two eigenvectors are linearly independent, meaning one cannot express the first in terms of the second or *vice versa*.

Note that there is a special shortcut to finding eigenvectors of Hermitian 2×2 matrices. Since the eigenvectors form an orthonormal set, if $e_+ = \begin{pmatrix} \alpha_+ \\ \beta_+ \end{pmatrix}$, then we immediately find that $e_- = \begin{pmatrix} \beta_+^* \\ -\alpha_+^* \end{pmatrix}$. Or, in other words, $\alpha_- = \beta_+^*$ and $\beta_- = -\alpha_+^*$.

We use eigenvalues and eigenvectors all the time in quantum mechanics. For 2×2 matrices, we need to recall that all Hermitian matrices have two real eigenvalues, the eigenvectors can be chosen to form an orthonormal set, and the eigenvectors are complete. Eigenvectors remain eigenvectors after multiplying the vector by an arbitrary nonzero number—if they are normalized, the arbitrariness is only a complex phase $e^{i\theta}$.

Finally, let us reiterate why we focus so much on Hermitian matrices. It turns out that most measurable quantities in quantum mechanics are associated with a Hermitian matrix (in a particular basis). It is often said this is *required* for all measurable quantities, because measurements always produce real numbers, but the situation is a bit more subtle than that, as we will see as we develop the quantum-mechanics formalism further.

Now, it is your turn for some practice with problems.

Section 5.7 Problems:

5.7.1 Eigenvalues

Find the eigenvalues of

$$A = \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 2 \\ 2 & 0 \end{pmatrix} \quad \text{and} \quad C = \begin{pmatrix} 1 & i \\ -i & -1 \end{pmatrix} \tag{5.154}$$

5.7.2 Eigenvalues and Eigenvectors

Find the eigenvalues and eigenvectors of

$$\begin{pmatrix} 0 & i \\ -i & 0 \end{pmatrix} \tag{5.155}$$

and verify that the eigenvectors you find are orthogonal.

5.7.3 Eigenvector Trick

Because eigenvectors of Hermitian matrices are orthonormal, we should have that if $e_+ = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$, then e_- must be $\begin{pmatrix} \beta^* \\ -\alpha^* \end{pmatrix}$, since that is the unit vector orthogonal to e_+ (recall that $*$ denotes the complex conjugate). Use this trick to find the eigenvalues and eigenvectors of

$$A = \begin{pmatrix} 2 & -1 \\ -1 & 1 \end{pmatrix}. \tag{5.156}$$

Verify that $Ae_+ = \lambda_+ e_+$ and $Ae_- = \lambda_- e_-$. Note that when using this trick, the second eigenvector may not have the first component being a real nonnegative number. This can be fixed, if desired, by multiplying the eigenvector by an overall complex phase to ensure the first component of the vector is a positive real number.

5.7.4 Completeness

Finish the derivation started in the text by calculating 2 2 matrix element of the outer product of the general eigenvectors to show that it is equal to 1.

5.7.5 Completeness Trick

Using the "trick" in Problem 5.7.3, let $e_+ = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$ and $e_- = \begin{pmatrix} \beta^* \\ -\alpha^* \end{pmatrix}$, with $|\alpha_+|^2 + |\beta_+|^2 = 1$. Use these results and these results only, to show that

$$\mathbb{I}_2 = e_+ \otimes e_+^\dagger + e_- \otimes e_-^\dagger \tag{5.157}$$

holds.

5.7.6 Hermitian Conjugate

Suppose that e_+ and e_- are eigenvectors of A with eigenvalues λ_+ and λ_-. Then

$$\sum_{j=1}^{2} A_{ij}(e_+)_j = \lambda_+ (e_+)_i \tag{5.158}$$

and

$$\sum_{j=1}^{2} A_{ij}(e_-)_j = \lambda_- (e_-)_i. \tag{5.159}$$

Show that the Hermitian conjugate equation also holds:

$$(e_+)^\dagger A = \lambda_+ (e_+)^\dagger \text{ and } (e_-)^\dagger A = \lambda_- (e_-)^\dagger, \tag{5.160}$$

by taking the Hermitian conjugate of the eigenvalue equations. Use the facts that $\lambda_\pm$ are real and the A matrix is Hermitian, so $A_{ji} = A^*_{ij}$. Note that if the eigenvectors are complex valued, then the left eigenvectors are *different* from the right eigenvectors, because of the complex conjugate.

Hint: if you find the abstract problem to be challenging, try some concrete examples like $A = \begin{pmatrix} 1 & 2 \\ 2 & -1 \end{pmatrix}$ and $B = \begin{pmatrix} 2 & i \\ -i & -2 \end{pmatrix}$ and then try the general case.

5.7.7 Eigenvectors and Complex Phases

Identify the eigenvectors that correspond to the same state (or equivalently the same ray). This means the vectors are identical up to multiplication by a complex phase $e^{i\theta}$ with real θ.

(a) $\begin{pmatrix} \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \end{pmatrix}$

(b) $\begin{pmatrix} \frac{1+i}{2} \\ \frac{1-i}{2} \end{pmatrix}$

(c) $\begin{pmatrix} \frac{i}{\sqrt{2}} \\ -\frac{i}{\sqrt{2}} \end{pmatrix}$

(d) $\begin{pmatrix} \frac{1}{2\sqrt{2}} + \frac{\sqrt{3}i}{2\sqrt{2}} \\ -\frac{1}{2\sqrt{2}} - \frac{\sqrt{3}i}{2\sqrt{2}} \end{pmatrix}$

(e) $\begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1-i}{2} \end{pmatrix}$

(f) $\begin{pmatrix} -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix}$

Hint: find the factor $e^{i\theta}$ to multiply by that makes the top component of the vector real and positive.

5.7.8 Decomposition to Find Eigenvalues

Consider a matrix of the form

$$A = \begin{pmatrix} E_1 & V \\ V^* & E_2 \end{pmatrix} \tag{5.161}$$

with E_1 and E_2 both real. Define the average $\bar{E} = \frac{E_1+E_2}{2}$ and the difference $\Delta = \frac{E_1-E_2}{2}$. Write the matrix as

$$A = \bar{E}\mathbb{I}_2 + \begin{pmatrix} \Delta & V \\ V^* & -\Delta \end{pmatrix} \tag{5.162}$$

From this decomposition, you should be able to immediately show that the eigenvalues are $\bar{E} \pm \sqrt{\Delta^2 + |V|^2}$. This decomposition can be employed on any Hermitian 2×2 matrix to quickly obtain the eigenvalues.

5.7.9 Scaling and Rotating

Consider the matrix:

$$A = \begin{pmatrix} 0 & 2 \\ 2 & 0 \end{pmatrix}. \tag{5.163}$$

Show by drawing some pictures that the vector $\begin{pmatrix} a \\ b \end{pmatrix}$ is mapped to $\begin{pmatrix} 2b \\ 2a \end{pmatrix}$ when it is multiplied by A. Use picture to show for many different choices of a and b the vector that is multiplied by A is both rotated and scaled. However, if you pick $a = b = 1$, or $a = -b = 1$, those vectors are only scaled, not rotated. Show as well that these eigenvectors are orthogonal and that they span the two-dimensional vector space.

5.7.10 Viète's Formulas

A student of mine, Tomasz Szymanski, pointed out to me that the two results $\lambda_+ + \lambda_- = a + d$ and $\lambda_+\lambda_- = ad - |b|^2$ for Hermitian matrices can be employed to simplify some of the torturous algebra we employed in computing eigenvectors and the completeness relations. Viète's idea is to never have to work with a square root in the algebraic manipulations. For example, the expression $\lambda_+^2 + \lambda_-^2$ can be rewritten using the add-zero trick as

$$\lambda_+^2 + \lambda_-^2 + 2\lambda_+\lambda_- - 2\lambda_+\lambda_- = (\lambda_+ + \lambda_-)^2 - 2\lambda_+\lambda_-. \tag{5.164}$$

Using Viète's formulas, we find that

$$\begin{aligned}(\lambda_+ + \lambda_-)^2 - 2\lambda_+\lambda_- &= (a+d)^2 - 2(ad - |b|^2) \\ &= a^2 + 2ad + d^2 - 2ad + 2|b|^2 \\ &= a^2 + d^2 + 2|b|^2. \end{aligned} \tag{5.165}$$

In this way, we avoid having to ever work with the square roots. Revisit one of the problems, such as computing α_{12} for completeness and rearrange the algebra using Viète's rules.

Do you find this approach to be simpler than the standard way?

As a historical note, Francois Viète was a French mathematician from the sixteenth century.

5.8 Dirac bra and ket Notation

We now introduce a commonly used notation for quantum mechanics, which was invented by Paul Dirac in 1939 Dirac (1939). It is called the "bra" and "ket" notation to associate a bra-ket (sometimes also called a bracket) with a quantum expectation value or measurement. Until you have seen and understood this notation, it looks quite foreign. But you need to always remember, it is just a notation, nothing more. There is no scientific content to it; it is just a convenience, which converts otherwise long expressions into shorter more manageable ones, without losing any content! Sounds great. Let's get started.

To begin, we focus on the problems we have just been working on—the eigenvectors of 2×2 Hermitian matrices. Recall we had two eigenvectors, which we wrote as

$$e_+ = \begin{pmatrix} \alpha_+ \\ \beta_+ \end{pmatrix} \text{ and } e_- = \begin{pmatrix} \alpha_- \\ \beta_- \end{pmatrix} = \begin{pmatrix} \beta_+^* \\ -\alpha_+^* \end{pmatrix}, \tag{5.166}$$

where the last result follows from the orthogonality of the eigenvectors. The Dirac "bra" and "ket" are just new notation for representing row and column vectors. So, we use a ket for column vectors, where, in this example, $e_+ = |+\rangle = \begin{pmatrix} \alpha_+ \\ \beta_+ \end{pmatrix}$ and $e_- = |-\rangle = \begin{pmatrix} \alpha_- \\ \beta_- \end{pmatrix}$. The ket notation is always of the form, vertical line, label, angle bracket. Next, the row vectors are given by "bras" (Hermitian conjugates of kets) as follows:

$$e_+^\dagger = (\alpha_+^* \;\; \beta_+^*) = \langle +|, \;\; e_-^\dagger = (\alpha_-^* \;\; \beta_-^*) = (\beta_+ \;\; -\alpha_+) = \langle -|. \tag{5.167}$$

Note how the bra has an angle bracket to the left, a label that describes the state in the center and a vertical bar on the right that "closes" the bra—this is like the mirror image of a ket.

We combine "bras" and "kets" as a bra-ket (dropping the double line to a single line) to describe inner (or dot) products as

$$\langle +|+\rangle = e_+^\dagger \cdot e_+ = (\alpha_+^* \;\; \beta_+^*) \cdot \begin{pmatrix} \alpha_+ \\ \beta_+ \end{pmatrix} = |\alpha_+|^2 + |\beta_+|^2 = 1$$

$$\langle +|-\rangle = e_+^\dagger \cdot e_- = (\alpha_+^* \;\; \beta_+^*) \cdot \begin{pmatrix} \beta_+^* \\ -\alpha_+^* \end{pmatrix} = \alpha_+^* \beta_+^* - \beta_+^* \alpha_+^* = 0. \tag{5.168}$$

Please work out for yourself what $\langle -|+\rangle$ and $\langle -|-\rangle$ are equal to. The idea of this notation is that the bra-ket looks like a "closed" object $\langle a|b\rangle$, which indicates that the final result is a scalar (complex) number. Note that a bra $\langle a|$ is the Hermitian conjugate of a ket $|a\rangle$, when we represent them as row and column vectors. The bra and ket notation works for any vector, not just for eigenvectors.

We can use this notation to indicate matrix-vector multiplication. For example, the eigenvalue-eigenvector relationship, given by $Ae_\pm = \lambda_\pm e_\pm$, translates into

$$\underbrace{\hat{A}}_{\text{operator}} \underbrace{|\pm\rangle}_{\text{state}} = \underbrace{\lambda_\pm}_{\text{number}} \underbrace{|\pm\rangle}_{\text{state}}. \tag{5.169}$$

Here, we used $\hat{A}$ ("A-hat") to denote the matrix A, but thought of now as an abstract operator. We describe what an abstract operator is in the next paragraph. Here, the symbol $\lambda_\pm$ is still the eigenvalue (real number) for the eigenvalue equation. The vector is now represented by the ket and we often call bras and kets states instead of vectors. The Hermitian conjugate equation is $\langle \pm|\hat{A} = \lambda_\pm \langle \pm|$, where we used the fact that $\hat{A} = \hat{A}^\dagger$, for a Hermitian operator (and the fact that eigenvalues of Hermitian operators are real). Note here that the operator acts to the left on the bra. In general, in Dirac notation, an operator can act to the left or to the right. This freedom can be viewed as a nice aspect of the notation, although mathematicians are bothered by the ambiguity that it brings, and for this reason, they do not like using Dirac notation. But don't worry, it won't get you into trouble here; we will point out all the subtleties for you as they occur.

We use the term abstract operator (often shortened to just operator) to indicate something that acts on a state (or ket) and maps it to a new state (or ket), multiplied by a possibly complex number. We can always think of representing abstract operators via matrices and the states as vectors, so that an operator acting on a state can be synonymous with a matrix multiplying a vector. In the bra-ket notation, operators can act to the right or to the left—this is like a matrix being multiplied from the left by a row vector or from the right by a column vector. If it is multiplied from the right and from the left, then we create a number again—in Dirac notation, we write $v^\dagger A v'$ as $\langle v|\hat{A}|v'\rangle$. You can think of this as a sandwich—the bra and ket are the "bread" and the operator is the "filling."

It is useful to take a little break here and recap what we have done. We use the Dirac bra and ket notation to indicate different vectors (or states). We use a hat to indicate an abstract operator (which can also be schematically thought of as a matrix), and we sandwich bras and kets together to indicate inner (dot) products between vectors (which is a complex number). It is just a more compact way of representing everything we have been working with so far.

So, for example, we can use this notation to describe the Stern-Gerlach states we worked with earlier. Let $|\uparrow\rangle$ denote a positive projection on the $+z$ axis, $|\downarrow\rangle$ is a negative projection on the $+z$ axis, $|\rightarrow\rangle$ is a positive projection on the $+x$ axis, and $|\leftarrow\rangle$ is a negative projection on the $+x$ axis. Then, $|\nearrow\rangle$ would indicate a positive projection on an axis at an angle θ to the vertical. One can see it is a little difficult to tell precisely what the angle θ is, so it is more common to use a notation where $|\uparrow\rangle$ denotes a positive projection and $|\downarrow\rangle$ denotes a negative projection. Then, we use a subscript to indicate the axis. So, we would have notations like $|\uparrow\rangle_z$ or $|\uparrow\rangle_x$ or $|\downarrow\rangle_\theta$ to indicate different states oriented on different axes. Using this notation, we have identities such as

$$|\uparrow\rangle_z = |\downarrow\rangle_{-z} \text{ and } |\downarrow\rangle_z = |\uparrow\rangle_{-z}, \tag{5.170}$$

which means, in words, the positive projection on the z-axis is the same as a negative projection on the minus z-axis, and so on.

Similarly, we can describe the probability amplitude on the screen for single and two-slit experiments via a generic symbol $|\psi\rangle$. You can think of $|\psi\rangle$ as the abstract set of rules we use to describe all alternative ways to get from the source to the detector, expressed generically as the rules, without evaluating them at a specific position of the detector. Because the amplitude arrow changes with position, we need to introduce a bra to indicate the position where we want the probability arrow to be evaluated for a specific detector. Then, $\langle x|\psi\rangle$ is defined to be $\psi(x)$, which is the probability amplitude (arrow) at position x and is given by applying the abstract rules to determine the final arrow after hooking all the specific arrows for the alternative ways to leave the source and arrive at the specific position x. This object $\psi(x)$ is called the *wavefunction*; it and its modulus squared, which is the *probability density*, are both plotted in Fig. 5.16. Note the difference between the quantum state, which is the set of rules, and the wavefunction, which involves evaluating those rules at a particular spatial location, resulting in a number (arrow). It is important to understand properly how these different objects are determined and what they mean.

Of course, we square the length of the arrows to determine the probability at x: $P(x) = |\psi(x)|^2 = |\langle x|\psi\rangle|^2$. The object $\psi(x) = \langle x|\psi\rangle$, which we call the wavefunction, is an inner product. It is the projection of the quantum state $|\psi\rangle$ onto the eigenvectors of position $\langle x|$. While it is the mainstay of essentially all quantum textbooks, we will not focus on it as much as they do, because we will not need it as much in the way we develop the quantum theory.

So, to summarize, the Dirac bra and ket notation is just a compact and elegant way of describing objects we have been working with already. It turns out to be a

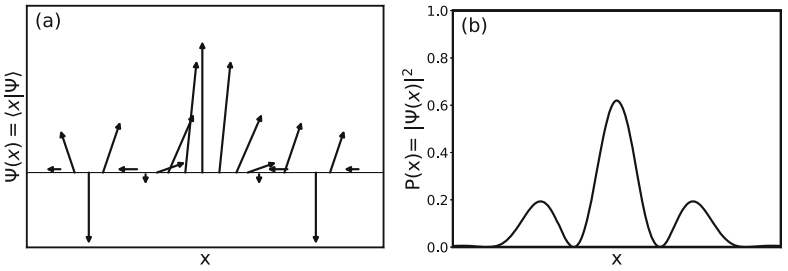


Fig. 5.16 Schematic for how the final arrows become the wavefunction (left) and how the square of the wavefunction becomes the probability density (right)

common language often used to describe quantum mechanics and is a notation we will use freely for the remainder of the book. We will develop its use slowly, to ensure you are comfortable using it.

Here is an executive summary of Dirac notation: a state $|\psi\rangle$ is a normalized state, if it satisfies $\langle\psi|\psi\rangle = 1$. The closed bra-ket $\langle\phi|\psi\rangle$ is always a complex number. An operator $\hat{A}$ maps a state to a state, which we denote as $\hat{A}|\psi\rangle$. Here, we include a hat to indicate that we can think of it as an abstract operator (representation independent). The product $\hat{A}|\psi\rangle$ is a ket. Similarly, the product $\langle\psi|\hat{A}$ is a bra. Then, $\langle\phi|\hat{A}|\psi\rangle$ is a complex number (also called a matrix element), because we "collapsed" a bra with a ket.

We have been discussing abstract operators and matrices and acting as if they can be simply interchanged. But they are different, so we want to carefully unpack them here. The difference is similar to the difference between a vector and the expansion of a vector in a particular basis. The abstract operator will map a vector into another vector. To define it, we simply need to say what the map is. At this stage, it is not a matrix. To represent it as a matrix, we need to convert from operators to numbers. The way we do this is to use our completeness relation (from the end of the previous section), written out in Dirac notation as $|e_1\rangle\langle e_1| + |e_2\rangle\langle e_2| = \mathbb{I}_2$, where $\{|e_1\rangle, |e_2\rangle\}$ is the orthonormal basis set, with $\langle e_i|e_j\rangle = \delta_{ij}$; δ_{ij} is the Kronecker delta symbol—it is 1 if $i = j$ and is 0 if $i \neq j$. In Dirac representation, we suppress the tensor product, so the ket-bra product is really analogous to the tensor product of the column vector with the row vector, which we saw led to a matrix, previously. In Dirac notation, we recognize that a ket-bra product is an operator, because it maps a ket vector to another ket vector. Continuing in the derivation, we take our original multiplication $\hat{A}|\psi\rangle$ and multiply by one to the right and the left of the operator:

$$\begin{aligned}
\hat{A}|\psi\rangle &= \mathbb{I}_2\hat{A}\mathbb{I}_2|\psi\rangle \\
&= \sum_i |e_i\rangle\langle e_i|\hat{A}|\sum_j |e_j\rangle\langle e_j|\psi\rangle \\
&= |e_1\rangle\langle e_1|\hat{A}|e_1\rangle\langle e_1|\psi\rangle + |e_1\rangle\langle e_1|\hat{A}|e_2\rangle\langle e_2|\psi\rangle \\
&+ |e_2\rangle\langle e_2|\hat{A}|e_1\rangle\langle e_1|\psi\rangle + |e_2\rangle\langle e_2|\hat{A}|e_2\rangle\langle e_2|\psi\rangle. \qquad (5.171)
\end{aligned}$$

All of the bra-kets are complex numbers. If we denote $\langle e_1|\psi\rangle = \alpha$ and $\langle e_2|\psi\rangle = \beta$, then we have the correspondence:

$$|\psi\rangle = |e_1\rangle\alpha + |e_2\rangle\beta \;\leftrightarrow\; \begin{pmatrix}\alpha\\ \beta\end{pmatrix}. \tag{5.172}$$

Note how the vector $|\psi\rangle$ is abstractly expanded in the basis $|e_i\rangle$ and how this is isomorphic to the standard column vector. It is often useful to think of the inner product formed by a bra $\langle\phi|$ and a ket $|\psi\rangle$ as determining how much of the ket $|\phi\rangle$ is "in" the ket $|\psi\rangle$—in other words, it is the *projection* of the ket $|\psi\rangle$ onto the ket $|\phi\rangle$. The matrix works in the same way. We denote $\langle e_1|\hat{A}|e_1\rangle = a$, $\langle e_1|\hat{A}|e_2\rangle = b$, $\langle e_2|\hat{A}|e_1\rangle = c$, and $\langle e_2|\hat{A}|e_2\rangle = d$. If the operator $\hat{A}$ is Hermitian, then we have $c = b^*$. This can be understood as follows. Since a bra-ket is an inner product, according to our rules, we have $\langle\phi|\psi\rangle = \langle\psi|\phi\rangle^*$. When we have operators in the bra-ket, complex conjugation of the bra-ket "sandwich" changes the operator in the "filling" to its Hermitian conjugate according to

$$\langle\phi|\hat{A}|\psi\rangle^* = \langle\psi|\hat{A}^\dagger|\phi\rangle. \tag{5.173}$$

Hence, we have the equivalence:

$$\hat{A} = |e_1\rangle\langle e_1|a + |e_1\rangle\langle e_2|b + |e_2\rangle\langle e_1|c + |e_2\rangle\langle e_2|d \;\leftrightarrow\; \begin{pmatrix}a & b\\ c & d\end{pmatrix}. \tag{5.174}$$

Recall how we said before that the outer product given by a ket-bra product is an operator, and here, you can see how using these outer products with a set of basis vectors allows us to completely specify the operator. In addition, the matrix elements of the Hermitian conjugate operator correspond to the complex conjugate transpose of the matrix representation of the original operator. Finally, one should see that Eq. (5.171) has the following equivalence:

$$\begin{aligned}\hat{A}|\psi\rangle &= \big(|e_1\rangle\langle e_1|a + |e_1\rangle\langle e_2|b + |e_2\rangle\langle e_1|c + |e_2\rangle\langle e_2|d\big)\big(|e_1\rangle\alpha + |e_2\rangle\beta\big)\\ &= |e_1\rangle\big(a\alpha + b\beta\big) + |e_2\rangle\big(c\alpha + d\beta\big)\\ &\leftrightarrow \begin{pmatrix}a & b\\ c & d\end{pmatrix}\begin{pmatrix}\alpha\\ \beta\end{pmatrix} = \begin{pmatrix}a\alpha + b\beta\\ c\alpha + d\beta\end{pmatrix}.\end{aligned} \tag{5.175}$$

I hope these correspondences are clear to you.

We end the section with a concrete description for how we go from an abstract operator to a matrix representation. Suppose the operator $\hat{A}$ is defined via taking the unit norm ket $|\phi\rangle$ and multiplying it by a and taking the orthogonal unit norm ket $|\psi\rangle$ and multiplying it by d. It then is represented by

$$\hat{A} = |\phi\rangle\langle\phi|a + |\psi\rangle\langle\psi|d. \tag{5.176}$$

Now, we can directly compute how $\hat{A}$ acts on a general vector given by $|\chi\rangle = \alpha|\phi\rangle + \beta|\psi\rangle$. We find

$$\begin{aligned}\hat{A}|\chi\rangle &= \hat{A}\big(\alpha|\phi\rangle + \beta|\psi\rangle\big) \\ &= \alpha\big(\underbrace{\hat{A}|\phi\rangle}_{a|\phi\rangle}\big) + \beta\big(\underbrace{\hat{A}|\psi\rangle}_{d|\psi\rangle}\big) \\ &= a\alpha|\phi\rangle + d\beta|\psi\rangle. \end{aligned} \tag{5.177}$$

Our second example considers the two vectors:

$$|\phi\rangle \leftrightarrow \frac{1}{\sqrt{5}}\begin{pmatrix} 2 \\ -1 \end{pmatrix} \text{ and } |\psi\rangle \leftrightarrow \frac{1}{\sqrt{5}}\begin{pmatrix} 1 \\ 2 \end{pmatrix}. \tag{5.178}$$

You can immediately verify that $\langle\phi|\phi\rangle = \langle\psi|\psi\rangle = 1$ and $\langle\phi|\psi\rangle = 0$. Suppose the operator $\hat{B}$ is represented by the matrix $\begin{pmatrix} 4 & -i \\ i & 2 \end{pmatrix}$. We want to change the basis from the current basis (given by the vectors $\left\{\begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix}\right\}$ into the $\phi - \psi$ basis. To do this, we need to compute the four matrix elements $\langle\phi|\hat{B}|\phi\rangle$, $\langle\phi|\hat{B}|\psi\rangle$, $\langle\psi|\hat{B}|\phi\rangle$, and $\langle\psi|\hat{B}|\psi\rangle$. Direct computation gives

$$\begin{aligned}\langle\phi|\hat{B}|\phi\rangle &= \frac{1}{5}(2 \quad -1)\cdot\begin{pmatrix} 4 & -i \\ i & 2 \end{pmatrix}\cdot\begin{pmatrix} 2 \\ -1 \end{pmatrix} = \frac{18}{5} \\ \langle\phi|\hat{B}|\psi\rangle &= \frac{1}{5}(2 \quad -1)\cdot\begin{pmatrix} 4 & -i \\ i & 2 \end{pmatrix}\cdot\begin{pmatrix} 1 \\ 2 \end{pmatrix} = \frac{4}{5} - i \\ \langle\psi|\hat{B}|\phi\rangle &= \frac{1}{5}(1 \quad 2)\cdot\begin{pmatrix} 4 & -i \\ i & 2 \end{pmatrix}\cdot\begin{pmatrix} 2 \\ -1 \end{pmatrix} = \frac{4}{5} + i \\ \langle\psi|\hat{B}|\psi\rangle &= \frac{1}{5}(1 \quad 2)\cdot\begin{pmatrix} 4 & -i \\ i & 2 \end{pmatrix}\cdot\begin{pmatrix} 1 \\ 2 \end{pmatrix} = \frac{12}{5}. \end{aligned} \tag{5.179}$$

We say that the operator $\hat{B}$ is given by the following in the $\{|\phi\rangle, |\psi\rangle\}$ basis:

$$\begin{pmatrix} \langle\phi|\hat{B}|\phi\rangle & \langle\phi|\hat{B}|\psi\rangle \\ \langle\psi|\hat{B}|\phi\rangle & \langle\psi|\hat{B}|\psi\rangle \end{pmatrix} = \begin{pmatrix} \frac{18}{5} & \frac{4}{5} - i \\ \frac{4}{5} + i & \frac{12}{5} \end{pmatrix}. \tag{5.180}$$

Note that in this new basis, the matrix remains Hermitian, just like the original matrix. In addition, the trace is 6 for the original and transformed matrix and the determinant is 7 for both as well. These are general results. The change of basis of a matrix preserves both the determinant and the trace (as well as special properties such as Hermiticity or unitarity). The operator is always referred to as $\hat{B}$ and is

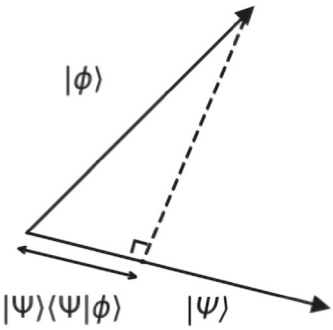


Fig. 5.17 Schematic of the projection of the vector $|\phi\rangle$ on $|\psi\rangle$. As can be seen from the double arrow highlighting $|\psi\rangle\langle\psi|\phi\rangle$, the projection (or, if you like, the inner product of $\langle\psi|\phi\rangle$) shows how much of $|\psi\rangle$ is in $|\phi\rangle$ (we assume both vectors are unit norm)

independent of the basis. Once a basis is chosen, we can compute the representation of the operator as a matrix in that basis.

I have found students often struggle with making sense of how to convert from an abstract operator to the matrix that represents it. It is an important task to master, so please take the time to understand it.

Section 5.8 Problems:

5.8.1 Projection Operator

The operator $\hat{P}_\psi = |\psi\rangle\langle\psi|$ is called the projection operator onto state $|\psi\rangle$ (see Fig. 5.17). Show that if $|\psi\rangle$ is normalized, then $\hat{P}_\psi^2 = \hat{P}_\psi$. Show that $\hat{P}_\psi|\phi\rangle$ projects the vector $|\phi\rangle$ onto the vector $|\psi\rangle$, meaning the result is the inner product of $|\psi\rangle$ with $|\phi\rangle$ multiplied by the vector $|\psi\rangle$, or $\left(\psi^\dagger \cdot \phi\right)\psi$ in vector notation.

5.8.2 Complex Conjugation in Dirac Notation

Since $\langle\phi|\hat{A}|\psi\rangle$ is a complex number, we can take its complex conjugate. The rule is that

$$\langle\phi|\hat{A}|\psi\rangle^* = \langle\psi|\hat{A}^\dagger|\phi\rangle, \tag{5.181}$$

where we interchange bra to ket and ket to bra and we take the Hermitian conjugate of the operator. Let

$$|\psi\rangle = \begin{pmatrix}\alpha\\ \beta\end{pmatrix}, \quad |\phi\rangle = \begin{pmatrix}\gamma\\ \delta\end{pmatrix}, \quad \text{and} \quad A = \begin{pmatrix}a & b\\ c & d\end{pmatrix}, \tag{5.182}$$

where all symbols are complex numbers. Compute $\langle\phi|\hat{A}|\psi\rangle$, take its complex conjugate, and show it is equal to $\langle\psi|\hat{A}^\dagger|\phi\rangle$.

Hint: recall if $|\phi\rangle = \begin{pmatrix}\gamma\\ \delta\end{pmatrix}$, then $\langle\phi| = (\gamma^* \;\; \delta^*)$.

5.8.3 Completeness

The completeness relation is written as

$$\mathbb{I} = |+\rangle\langle+| + |-\rangle\langle-|, \tag{5.183}$$

which is the sum of the projection operators corresponding to the $|+\rangle$ and $|-\rangle$ states. Use $|+\rangle = \begin{pmatrix}\alpha\\ \beta\end{pmatrix}$ and $|-\rangle = \begin{pmatrix}\beta^*\\ -\alpha^*\end{pmatrix}$, and show that

$$\big(|+\rangle\langle+| + |-\rangle\langle-|\big)|\phi\rangle = |\phi\rangle, \tag{5.184}$$

for $|\phi\rangle = \begin{pmatrix}\gamma\\ \delta\end{pmatrix}$, by converting the bra and ket notation to 2×2 matrices and performing the matrix multiplications and sums. Recall that $|\alpha|^2 + |\beta|^2 = 1$.

Hint: $|+\rangle\langle+| = \begin{pmatrix}\alpha\\ \beta\end{pmatrix} \otimes (\alpha^* \;\; \beta^*)$, the outer product.

5.8.4 Representation of an Operator as a Matrix in a Specific Basis

Let $|+\rangle$ and $|-\rangle$ be the eigenvectors of $\hat{A}$, which is represented by a 2×2 Hermitian matrix with eigenvalues λ_+ and λ_-. Compute

$$\langle+|\hat{A}|+\rangle, \;\; \langle+|\hat{A}|-\rangle, \;\; \langle-|\hat{A}|+\rangle, \;\; \langle-|\hat{A}|-\rangle. \tag{5.185}$$

Assume $\langle+|+\rangle = \langle-|-\rangle = 1$ and $\langle+|-\rangle = 0$. Your answers should depend only on λ_+ and λ_-.

5.8.5 Eigenvectors of the Projection Operator

Suppose $\hat{P}_+ = |+\rangle\langle+|$ is the projection operator onto the $|+\rangle$ eigenstate. Show that $|+\rangle$ and $|-\rangle$ are eigenvectors of $\hat{P}_+$ with eigenvalues 1 and 0, respectively.

5.8.6 Ray Versus Vector

We have mentioned already that multiplying a vector by a nonzero complex number does not change the quantum state. In essence, just the direction of the vector is important, not its length or its complex phase. To show how this enters into calculations, one often writes a quantum state in an operator form (also called a density matrix) that is denoted by ρ_ψ and is given by

$$\rho_\psi = \frac{|\psi\rangle\langle\psi|}{\langle\psi|\psi\rangle}. \tag{5.186}$$

Show that if you define a new state from a rescaled vector $|\phi\rangle = c|\psi\rangle$, that the density matrix is unchanged. That is, show that we have $\rho_\psi = \rho_\phi$ for $c \neq 0$.

5.8.7 Nondegenerate Eigenvectors of a Hermitian Operator Must Be Orthogonal

In this problem, we will consider a general case, rather than just 2×2 matrices. Suppose we have a set of eigenvectors $\{|\psi_n\rangle\}$ that satisfy

$$\hat{O}|\psi_n\rangle = \lambda_n|\psi_n\rangle \tag{5.187}$$

for a Hermitian operator $\hat{O}$, with λ_n the (real-valued) eigenvalues. We assume we are in the *nondegenerate* case, where $\lambda_n \neq \lambda_m$ for all $m \neq n$.

Show that $\langle\psi_n|\psi_m\rangle = 0$ for $m \neq n$.

Hint: consider evaluating $\langle\psi_m|\hat{O}|\psi_n\rangle$ two different ways for $n \neq m$ by operating the operator to the left or to the right. Use the fact that these states are eigenvectors and that the eigenvalues are different to conclude the orthogonality of the vectors. Work abstractly using only what you know.

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